

DEVELOPMENT AND USABILITY EVALUATION OF AN INTERACTIVE DATA STORY FOR SELECTED MOBILITY INFORMATION

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ABSTRACT

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In contemporary digital ecosystems, the accessibility of complex mobility data remains a significant challenge for non-technical audiences. While Deutsche Bahn (DB) and similar entities generate vast quantities of spatio-temporal data, existing dashboards often suffer from static presentations that limit user engagement and insight discovery. This thesis explores the integration of data storytelling with interactive dashboard design to bridge this gap by developing a prototype dashboard that incorporates varying degrees of narrative guidance and interactive freedom to examine how these elements influence user exploration, comprehension, and time-to-insight. To ground the design process in real user needs, we conducted exploratory focus groups to capture expectations, pain points, and preferred storytelling strategies, and translated these insights into the iterative development of a single prototype. The prototype was subsequently evaluated through structured usability testing sessions in which participants completed predefined tasks while thinking aloud, enabling us to assess usability, effectiveness, and the impact of narrative elements on understanding. Findings from both stages indicate a strong user preference for “balanced” dashboard models that combine narrative scaffolding with exploratory freedom, providing the basis for empirically grounded design guidelines for next-generation mobility dashboards.

ZUSAMMENFASSUNG:

In modernen digitalen Ökosystemen stellt die Zugänglichkeit komplexer Mobilitätsdaten für ein nicht-technisches Publikum nach wie vor eine große Herausforderung dar. Während die Deutsche Bahn (DB) und ähnliche Unternehmen riesige Mengen an räumlich-zeitlichen Daten generieren, leiden bestehende Dashboards oft unter statischen Darstellungen, die das Engagement der Nutzer und die Gewinnung von Erkenntnissen einschränken. Diese Arbeit untersucht die Integration von Data Storytelling in das Design interaktiver Dashboards, um diese Lücke zu schließen. Dazu wurde ein Prototyp-Dashboard entwickelt, das unterschiedliche Grade an narrativer Führung und interaktiver Freiheit beinhaltet, um zu untersuchen, wie diese Elemente die Erkundung, das Verständnis und die Zeit bis zur Erkenntnisgewinnung der Nutzer beeinflussen. Um den Designprozess auf die tatsächlichen Bedürfnisse der Nutzer auszurichten, führten wir explorative Fokusgruppen durch, um Erwartungen, Schwachstellen und bevorzugte Storytelling-Strategien zu erfassen, und setzten diese Erkenntnisse in die iterative Entwicklung eines einzigen

Prototyps um. Der Prototyp wurde anschließend durch strukturierte Usability-Tests evaluiert, in denen die Teilnehmer vordefinierte Aufgaben unter lautem Nachdenken ausführten, sodass wir die Benutzerfreundlichkeit, die Effektivität und den Einfluss narrativer Elemente auf das Verständnis bewerten konnten. Die Ergebnisse beider Phasen zeigen eine starke Präferenz der Benutzer für „ausgewogene“ Dashboard-Modelle, die narrative Strukturen mit explorativer Freiheit kombinieren und die Grundlage für empirisch fundierte Designrichtlinien für Mobilitäts-Dashboards der nächsten Generation bilden.

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INTRODUCTION

In contemporary digital ecosystems, organizations and institutions generate unprecedented volumes of data daily. The proliferation of sensors, IoT devices, and automated logging systems has created a “data deluge” that presents both opportunities and significant challenges. While raw data serves as a critical resource for strategic decision-making, its utility is fundamentally constrained by the limits of human cognition and the challenges of accessibility [1]. For non-technical audiences such as daily commuters or public transport policy observers raw datasets are opaque and unintelligible without appropriate mediating layers of abstraction and visualization [6]. Effective communication of data is, therefore, not merely a technical task of rendering pixels, but a complex cognitive and design challenge. It requires bridging the semantic gap between the statistical properties of a dataset and the mental models of the user [32].

1.1 MOTIVATION AND BACKGROUND

A growing solution to this communication challenge is **data storytelling**. This approach integrates the rigors of data visualization with the engaging structures of narrative to transform complex datasets into accessible stories [1]. Unlike static visualizations or standard exploratory dashboards which often leave interpretation entirely to the user, potentially leading to “**analysis paralysis**”, data stories guide audiences through insights, systematically highlighting patterns, contextualizing findings, and explaining causal relationships [25][39]. Scholars such as Dykes [11] and Hullman and Diakopoulos [22] argue that narrative framing is essential for insight retention. By imposing a narrative structure (a beginning, middle, and end), data storytellers can reduce cognitive load and focus user attention on critical anomalies or trends [40]. When combined with interactivity, data storytelling evolves into **narrative visualization** [39], empowering users to explore data dynamically, tailor narratives to their specific needs, and foster a deeper sense of engagement [4] [23].

The motivation for this thesis stems from the observation that while mobility datasets are increasing in scale, the tools available to the public to understand them are not evolving at the same pace [2]. This disparity highlights a crucial gap in current visualization practice, leading to the central focus of this research: the optimal balance between guided narrative and free exploration in data dashboards.

The **limitations of static dashboards** are clear: while these traditional tools communicate information quickly, they often fail to support deep exploratory analysis or personalized discovery [4]. Static

dashboards present the “what” but rarely explain the “why,” thus limiting the user to pre-determined conclusions [9]. Conversely, the **potential of interactivity** is vast, allowing users to filter, drill down, and follow multiple narrative paths, fostering active learning and deeper sense-making [12][44]. However, this freedom comes at a cost, as unguided interactivity can be overwhelming, leading to information overload and potentially frustrating the user. Therefore, this study hypothesizes that a **hybrid approach** combining the structured guidance of storytelling with the dynamic freedom of interactive exploration can significantly improve dashboard usability and comprehension for mobility data [7][40]. The project utilizes real-time mobility datasets to investigate this optimal balance.

1.2 PROBLEM STATEMENT

The specific problem addressed by this thesis resides in the application of visual analytics to the domain of mobility and transportation. Mobility data is uniquely complex; it is inherently spatio-temporal, high-velocity, and multivariate [2]. Understanding a transport network involves grasping the intricate interplay between time (delays, schedules), space (geographical routes, regional impacts), and magnitude (passenger volume, congestion severity) [41]. The primary subject of this research is the operational data environment of Deutsche Bahn (DB). As a leading mobility provider, DB generates massive, continuous streams of operational data [10][15]. However, despite the richness of this underlying data, the current user-facing interfaces, available through DB APIs and internal analytics systems [28][30], often lag in usability. Existing reports and public dashboards tend to be static, tabular, or strictly operational, lacking the personalized narrative layer that helps a commuter or decision-maker answer the critical question: “How does this affect me?”

The most critical deficiency is the absence of narrative context. Raw data and abstract visualizations describe **what** is happening, but they rarely explain **why** it matters or **how** it relates to the broader system. Current systems function more as data repositories than effective communication tools, lacking the “connective tissue” of storytelling annotations, guided sequences, and contextual highlights that help users build a mental model of the data[39]. For a layperson, a delay heat map is an abstract visualization; a narrative that guides them through “How winter storms affected the Hamburg-Berlin line” transforms that abstraction into actionable knowledge. The lack of this explanatory scaffolding restricts sense-making and leaves users to decipher complex statistical phenomena without guidance [22]. This research aims to overcome these deficits by empirically testing a hybrid visualization approach.

1.3 RESEARCH QUESTIONS, METHODOLOGY AND STRUCTURE

This study investigates the design space of interactive mobility dashboards, seeking to determine how narrative elements can be effectively combined with interactive affordances to enhance usability. Specifically, the research addresses the following overarching question:

How can a mobility data dashboard be developed into a data story to improve user understanding and increase their motivation to explore data?

To address this primary question, the study employs a systematic approach focused on the critical aspects of interactive data storytelling:

- **User Exploration and Interactivity:** Investigating how the introduction of interactive features influences user exploration behaviors in mobility data compared to static narratives.
- **Data Comprehension:** Determining whether guided storytelling improves data comprehension and reduces the time-to-insight for non-expert users, such as commuters.
- **Narrative Balance:** Identifying the optimal balance between author-driven narrative guidance and user-driven exploration in the mobility data context.

To address these points, a user-centered design process was employed. The primary objectives of this study are to **develop prototype dashboards** using real-world Deutsche Bahn data, to rigorously **evaluate the usability of interactive storytelling** within these prototypes, and finally, to **analyze user feedback** to recommend actionable design practices for mobility operators seeking to improve data transparency and communication.

The structure of this report is organized into six chapters.

Chapter 1, includes the **Introduction** part as being discussed here.

Chapter 2, **Literature Review**, first synthesizes existing research on data storytelling, narrative visualization, and the application of visual analytics to spatio-temporal mobility data.

Chapter 3, **Focus Group**, details the preparation, implementation, and evaluation of two focus group sessions conducted to gather qualitative user insights for the early conceptual design of the mobility dashboard prototype. The preparation involved selecting the focus group method for its ability to generate spontaneous feedback and choosing two contrasting public dashboards (Belgian Mobility and German Klima) for a comparative evaluation. The implementation describes the structured procedure, which involved a warm-up, guided exploration of the two existing dashboards, group comparison, and a creative closing activity where 11 student participants sketched their ideal dashboard. The evaluation integrates the findings, revealing critical design themes such as the importance of clear storytelling flow,

the need for essential interactivity (like filters and search), a preference for visual clarity, and the necessity of explanatory context to support data interpretation.

Chapter 4, **Prototype and Implementation of the Dashboard**, describes the conceptualization, design, and technical implementation of the high-fidelity dashboard prototype. The concept was inspired by the Belgian Mobility Dashboard and refined using the qualitative feedback gathered in the focus group, leading to the central idea of a narrative-driven data platform. The initial design used Figma for wireframes, but the need to test dynamic interactions necessitated building a functional application. The implementation leveraged a modern technology stack, including Next.js, React, TypeScript, Tailwind CSS, and Leaflet with OpenStreetMap for geospatial features. The final web-based prototype is structured around five interactive data stories: Punctuality, Congestion, Incidents, Multi-Modal Options, and Environmental Impact, which integrate features like an interactive map and a quiz to fulfill the project's requirement for interactive storytelling.

Chapter 5, **Usability Evaluation**, documents the execution and analysis of the usability test for the high-fidelity prototype. The primary goal was to evaluate the dashboard's effectiveness in data storytelling, focusing on user understanding, clarity, motivation, and recognition of the narrative. A mixed-methods approach was used, combining **Thinking Aloud protocols**, moderator observation, interviews, and the standardized **User Experience Questionnaire (UEQ)** [7]. The main test involved **seven participants** in a controlled lab environment. The quantitative UEQ results were highly positive, achieving an "Excellent" rating in Attractiveness. Qualitative analysis confirmed smooth interaction and high engagement with the interactive features, leading to the conclusion that the narrative approach was highly successful. The chapter concludes with minor suggestions for refinement and outlines a plan for future work involving live data integration to move the tool beyond static snapshots.

Finally, Chapter 6 provides the **Summary**, summarizes the study's two-phase methodology. The project successfully demonstrated that integrating storytelling principles into a mobility dashboard enhances user comprehension and engagement. The findings provide actionable design recommendations for improving data transparency in the mobility sector.

LITERATURE REVIEW

This literature review synthesizes the theoretical and practical literature relevant to the usability of interactive data stories. The focus is specifically on applications in real-time mobility dashboards, such as those prototyped for Deutsche Bahn [42][15].

To gather sources, the search strategy targeted peer-reviewed journal articles, major conference papers (from venues like ACM and IEEE), and foundational books. The search was focused on the last decade (2014–2025) but includes key earlier works to provide conceptual anchoring. Keyword searches combined terms such as “narrative visualization”, “data storytelling”, “dashboard usability”, “mobility dashboard”, and “visual analytics and storytelling”.

The review is organized thematically to build a logical argument. It first covers the theoretical foundations of data stories, including definitions and taxonomies [4][25]; it then examines interactivity and exploratory analysis [12][21] and the related cognitive and usability considerations [22][23]. Following this, the review addresses evaluation methods for these systems [1][7], discusses authoring tools and automation [19][23], and concludes by examining literature specific to spatio-temporal mobility visualization [2][8][9].

2.1 DEFINITION OF DATA STORIES

The foundation of any data story is the data itself [6][11]. The storytelling process begins with the selection, cleaning, and filtering of relevant data to uncover insights or support a specific message [31]. In the mobility domain, this data is uniquely complex, often characterized by its high volume, high velocity, and critical spatio-temporal components [2][41]. The primary goal of a data story is to translate this raw, often complex data into a structured, understandable, and engaging narrative [25][39].

A data story, often discussed under the term “narrative visualization,” is a communication method that combines data, visual elements, and narrative structures to convey insights [39]. It differs from a simple collection of charts by being a structured sequence of visualizations, deliberately designed to guide an audience to a specific conclusion or understanding [22]. The main objective is to frame data-driven insights in a compelling context, making complex information more engaging and easier to remember [6][11][23].

A Mobility Data Story (MDS) is a specific application of this concept, focused on spatio-temporal mobility data such as the movement of vehicles, people, or goods [2][41].

While it uses the same core principles of narrative [25], the primary difference is the nature of the data. Mobility data's large scale and complex spatial and temporal relationships [8] require specialized visualization techniques, such as animated maps, flow diagrams, or other visualizations capable of handling spatiotemporal information [17][41]. The narrative in an MDS aims to explain movement patterns, transportation system performance, or urban dynamics, typically for stakeholders in urban planning or logistics [2][8][9].

2.2 THE DEVELOPMENT PROCESS OF DATA STORY

The literature shows that constructing a data story is a multi-stage design process [29][31]. This process involves several key, often-overlapping, components:

- **Topic/Purpose:** The first step is defining a clear purpose [4][11]. The author must decide if the goal is explanatory (to communicate a specific finding) or exploratory (to facilitate sense-making) [25]. This purpose directly influences the intended cognitive effects, such as improving comprehension [7][40], enhancing information recall, or accelerating insight discovery [23]. The use of narrative framing and rhetorical devices is a key technique for guiding the audience's interpretation [22].
- **Data and Visualizations:** This involves the foundational steps of data preparation and the selection of appropriate visualizations [6][31]. The chosen visual encodings must effectively represent the data [20] and, in mobility contexts, must be able to handle complex spatio-temporal data [2][17].
- **Storytelling (Narrative Structures):** This is the core of the data story, where narrative structures are integrated. This involves choosing a narrative genre, such as a linear, author-driven path or a more flexible, reader-driven, interactive one [39].
- **Interaction:** Interaction is what elevates a data story from a static report. It provides mechanisms for the user to control the narrative's pace, filter data, or explore details [12][21][44]
- . A central design challenge is balancing the level of authorial guidance with the degree of user-driven exploration [25].
- **Evaluation:** Evaluating the story's effectiveness is a critical, and often difficult, final step [1]. Researchers and practitioners evaluate narrative visualizations by measuring outcomes like comprehension, time-to-insight [7][40], information recall [23], and user engagement [1]. These evaluation methods must account for the wide spectrum of storytelling, which now spans from explanatory slideshows to complex, interactive, multi-path experiences [1][39].

- **Automation:** Finally, recent technological trends focus on the automation of this development process [38]. This includes authoring tools that provide visualization recommendations [23] and new research leveraging large models to help generate the narrative text itself [19].

In the following sections, those elements will be explained further.

2.2.1 *Topic and Purpose of a Data Story*

The foundational stage of data storytelling is the explicit definition of its purpose [4][11]. The author must move beyond mere data presentation to articulate the specific analytical goal, which typically falls on a spectrum from purely *explanatory* to purely *exploratory* [34]. Explanatory stories aim to communicate a specific, pre-defined insight or finding, often leading to a linear narrative structure. This purpose directly influences the intended cognitive effects, such as improving comprehension [7][40], enhancing information recall, or accelerating insight discovery [23]. In contrast, exploratory systems are designed to facilitate user-driven sense-making, where the purpose is discovery [31]. For mobility data, the purpose might be descriptive (e.g., showing current delays), diagnostic (explaining the cause of a past congestion event), or predictive (forecasting future network performance). This initial definition is critical because the story's purpose dictates the subsequent choices regarding visualization type, narrative structure, and required interactivity [4]. The effective use of narrative framing and rhetorical devices, such as visual highlighting or annotations, is a key technique for guiding the audience's interpretation toward the author's intended purpose [34].

2.2.2 *Data and Visualization*

Spatio-Temporal Data and Visualization Challenges in Mobility will throw light on the specific domain of this project mobility that introduces unique and significant challenges, rooted in the nature of its data. Mobility datasets including their characteristics are distinguished by properties that drive specific design requirements:

1. **Spatio-temporal structure:** Events are tied to specific places and times [2].
2. **Heterogeneity:** Data comes from diverse sources such as sensors, ticketing systems, and schedule databases [41].
3. **High velocity:** Updates occur in near real-time, requiring dynamic visualization capabilities [41].
4. **Multiscale semantics:** Insights must be scalable from individual trips to network-wide summaries [2].

These properties create unique visualization constraints: enabling fluid temporal aggregation, maintaining geographic accuracy while avoiding map clutter, communicating uncertainty, and ensuring interactive performance at scale. Studies surveying urban mobility visualization recommend coordinated multiple views (map + timeline), temporal brushing, and flow maps, emphasizing that interactive filtering controls are essential [2][41].

Implications for Dashboards: For Deutsche Bahn-style mobility dashboards, this means: (1) map-centric views require focus + context techniques; (2) timeline controls must support fast re-aggregation; and (3) annotations must expose provenance and aggregation choices. These implications recur across domain frameworks for mobility dashboard design [8][9].

2.2.3 *Storytelling (Narrative Structures)*

Narrative visualization systems are typically characterized by where they fall on the continuum between author control and reader control [39]. On the author-driven side, the structure is often linear, with limited interaction, resembling a slideshow or a guided tour. This format excels at ensuring the audience receives a specific, clear message and improves comprehension for the intended insight [40]. However, it sacrifices the user's ability to verify or explore secondary insights. On the reader-driven side, the system offers maximal interactivity, allowing free exploration but potentially leading to a fragmented understanding or "analysis paralysis" [25]. Effective data stories often seek a middle ground, employing structures that use explicit narrative paths (e.g., step-by-step sequences) combined with interaction that allows for "escape" to a free exploration mode, and then a return to the narrative flow [1]. This hybrid structure aims to retain the cognitive benefits of narrative while providing the empowerment of exploration, balancing the user's desire for guidance with their need for validation and deeper understanding.

2.2.4 *Interaction*

Interaction is a foundational component of modern visualization, providing the mechanism for user-driven exploration and analysis [21][44]. It transforms the user from a passive observer into an active participant, enabling critical data manipulation tasks like filtering, selecting, and drilling down [12][21]. In the context of data stories, interaction plays a dual role: it can either control the pacing of the author's narrative or allow the reader to break away for independent exploration [39].

The taxonomy by Yi et al. remains a seminal reference for categorizing core interaction techniques in information visualization, encompassing actions such as select, explore, reconfigure, encode, abstract/elaborate, filter, and connect [44]. This taxonomy frames

interaction in terms of user intent and is highly relevant for designing standard dashboard controls (filtering, zooming, focus + context, linked views).

Later works extend these concepts by placing “**story control**” the navigation through narrative steps, branching, and bookmarks as a first-class interaction category in narrative visualization systems [39]. For designers of mobility dashboards, this necessitates considering both low-level manipulations (temporal aggregation, region selection) and higher-level narrative interactions (play/pause story, jump to annotated insight, restore author’s viewpoint) [8][9].

A consistent finding across empirical studies is that interactivity improves exploratory performance when it is purposeful and constrained; conversely, excessive interactivity or poorly signposted controls can increase cognitive load and reduce task efficiency [1]. Consequently, hybrid designs that combine a lightweight authorial narrative (framing, callouts, suggested filters) with targeted interactive capabilities tend to perform well on both comprehension and satisfaction metrics [7][40]. These hybrids are promising candidates for balancing engagement with usability, a core principle of user-centered design [32].

2.2.5 *Evaluation*

Given the complex interplay of narrative, interaction, and cognition, the final piece of the puzzle is evaluation. The literature provides a range of methods for assessing the effectiveness of these systems, moving from simple usability checks to measuring complex outcomes like insight generation and comprehension. Studies evaluating narrative visualizations and dashboards commonly combine objective performance metrics, such as task success, error rate, and time-to-insight [7][40], with standard usability measures like the User Experience Questionnaire (UEQ), and qualitative assessments including think-aloud protocols and interviews [1][13]. Recent systematic reviews of dashboard usability recommend mixed-methods approaches to capture both efficiency and the subjective qualities that shape long-term adoption, such as perceived usefulness, clarity, and trust [1]. Time-to-insight and “discovery” metrics are particularly appropriate for exploratory dashboards [7][40]; standardized scales remain useful for cross-study comparison [1][13].

To complement these quantitative measures, several recent efforts have attempted to adapt heuristic evaluation and checklists specifically to narrative visualizations [1][13]. Practitioner-led heuristics show promise as rapid formative evaluation methods [13]. However, the literature also documents that domain-specific criteria for mobility including spatial accuracy, latency, and temporal aggregation are necessary additions to general visualization heuristics [8][9]. Recent empirical work validating narrative visualization heuristics recommends combining these domain heuristics with standard usability

metrics for robust evaluation [13]. Furthermore, empirical studies that directly compare explanatory visualizations, interactive visualizations, and interactive data stories show mixed but informative results. Data stories often improve comprehension and recall for author-driven tasks and increase engagement [23][40]. Interactivity, on the other hand, enhances exploratory discovery but can reduce efficiency if not guided [7][18]. Recent controlled experiments indicate that a balanced approach (a guided story combined with targeted interactive affordances) can produce high task success, lower time-to-insight, and increased subjective satisfaction [7][40]. These cross-study consistencies provide a strong theoretical rationale for testing such balanced experimental conditions.

2.2.6 *Tools and Automation*

Following the review of evaluation methods, it is also important to consider the tools used to construct these systems. The landscape of authoring tools directly influences the design possibilities, trade-offs, and the potential for automation.

A recent mapping of data-story authoring tools distinguishes between (a) exploratory tools (e.g., Tableau, Kibana), (b) authoring frameworks for narrative sequencing, and (c) automated storytelling systems that synthesize narratives from data [38]. Practitioner platforms continue to converge: many modern Business Intelligence (BI) tools now support annotations, bookmarks, and lightweight narrative flows, while specialized “narrative” tools add storyboards and branching. Selecting tools for prototypes therefore entails trade-offs between speed of development, fidelity of narrative control, and support for interactive data exploration [31][32].

Automated approaches using AI to story generation have advanced rapidly [38]. Recent work explores using large language models (LLMs) and multimodal pipelines to propose candidate narratives, annotate visualizations, and generate textual explanations [19]. This can increase author productivity but also raises concerns about veracity and the need for human oversight. Other tools focus on automating the exploratory phase by providing visualization recommendations [18][44]. For mobility dashboards, these approaches could accelerate the generation of passenger-facing narratives but require careful validation to avoid misleading claims [19].

2.2.6.1 *Technology and Performance: Streaming, Latency, and Tool Trade-offs*

The technical implementation of a mobility dashboard introduces its own set of practical trade-offs. Modern mobility dashboards often rely on streaming pipelines, columnar storage, and visualization platforms that balance expressiveness with interactivity. Tool surveys and case studies highlight three practical trade-offs:

1. **Authoring Speed vs. Control:** BI tools (like Tableau or Kibana) accelerate prototype construction but constrain custom narrative behaviors. Custom web stacks (like D3 + React) provide maximal control but require more engineering resources [38].
2. **Authorial Narrative Support vs. Branching Exploration:** Many off-the-shelf tools support simple annotations, but advanced branching narratives still require customized frameworks. For a project comparing different story/interaction prototypes, this tool choice strongly affects feasibility [1][38].
3. **Realtime Performance vs. Complex Analytics:** Streaming dashboards that require sub-second UI response often simplify analytics or rely on precomputed aggregates, which can limit exploratory depth. This demands careful precomputation strategies, especially for high update rates [15].

2.3 MOBILITY DASHBOARDS

The general principles of data storytelling and dashboard design must be adapted to the specific challenges of the mobility domain. This context is defined by the unique characteristics of its data and the high-stakes needs of its users. Specifically, this research advocates for **mixing the approach of storytelling into dashboards** to enhance data comprehension for diverse stakeholders. Mobility data are typically spatio-temporal, high-frequency, and heterogeneous (e.g., passenger counts, delays, GPS traces) [2][41]. Visualizations must thus support multiscale temporal aggregation, geographical mapping, and fast filtering for real-time exploration [2][17]. Surveys of the field highlight the necessity for map-based interactions and coordinated multiple views to manage this inherent complexity[2][41].

Domain-specific design frameworks address these challenges by emphasizing (a) contextualization (geography and time), (b) purpose alignment (operational vs. public), and (c) performance scaling [8][9]. For a national rail operator such as Deutsche Bahn [10][15], these frameworks highlight the need to account for diverse stakeholders (commuters, operations managers, planners) and the policy implications of displayed metrics [8][9]. This recommendation highlights why the current study's focus on both commuters and data-literate users is necessary for robust evaluation. Furthermore, because public transport dashboards carry civic implications, the literature on trust

and rhetoric is highly relevant. Passengers infer service reliability from these dashboards. Visualization rhetoric research warns that framing choices (baselines, aggregation, color) can alter the perception of service quality [22]. The literature emphasizes transparent labeling (data source, timestamp, aggregation method) and communicating uncertainty to preserve trust [6]. For rail operators, over-simplified summaries risk eroding trust if they mask variability; accordingly, narrative elements should be used to contextualize rather than obscure raw data [6][22].

Case studies and real-world deployments from transport providers demonstrate the active use of BI and dashboarding solutions for internal operations and planning [10][15][34]. Public-facing dashboards from research and civic institutions, such as the C2SMART Mobility Data Dashboard (NYU) [5] and projects by the Urban Institute [42], illustrate the value of highly interactive, map-centric tools. Industry descriptions show DB's investment in data platforms for internal operations and productization (Mobimeo) [30]. However, public-facing narrative storytelling for commuters remains comparatively limited, representing a clear design opportunity. These examples, and similar public-sector deployments, often prioritize clarity and simplicity for civic audiences, using authored narratives on landing pages with optional interactive exploration[42]. Critically, existing deployments repeatedly flag mobile usability and information density as pain points [5][42]. For instance, the focus group's findings that one dashboard showed strong story clarity but poor performance, while the other offered high interactivity but suffered from clutter and poor mobile usability align with these broader cross-case findings in the literature. Narrative clarity tends to improve comprehension but is vulnerable to fragmentation [1][39]. Conversely, feature-rich interactive dashboards are engaging for power users but frequently overwhelm novices and mobile viewers [12][31]. Comparative deployment reports consistently recommend the hybrid, responsive designs this project investigates [8][9]. The corroborating design lessons from the literature support four concrete prescriptions for interactive storytelling in mobility dashboards: (1) provide a clear narrative entry point; (2) enable on-demand detail (e.g., tooltips); (3) limit information density on mobile; and (4) maintain consistent interaction metaphors to reduce cognitive load [1][8][32].

2.4 USABILITY CONSIDERATIONS OF DATA STORIES AND DASHBOARDS

While defining the *what* (narrative) and the *how* (interactivity) is crucial, the literature also emphasizes understanding the effect on the user. This involves analyzing the cognitive, rhetorical, and usability factors that determine whether a data story or dashboard is ultimately successful. Narrative devices (textual annotation, highlighting, sequencing) influence attention allocation and memory encoding [23]; visual rhetoric (framing, omission, emphasis) substantially shapes users' interpretations [22]. Cognitive theories applied in visualization research explain why narrative scaffolds and attention cues (visual salience, spotlighting) can reduce search time and improve recall [18]. However, this is only true if they align with task goals and do not overload perception with competing signals [31]. The concept of "visualization rhetoric" emphasizes that narrative visualizations are inherently persuasive: design choices encode editorial frames that can bias interpretation [22]. This is critical for mobility dashboards presenting delay statistics or congestion severity, as framing can lead to different user inferences about service quality [2][41]. Designers must therefore be explicit about data provenance, aggregation choices, and uncertainty to preserve trust and avoid misleading narrative framing [6].

To further ground the synthesis, this section explains empirical users studies on narrative visualization which provides a deeper review of the specific empirical evidence for the effects of narrative and interactivity on user performance and perception. Controlled experiments comparing static, interactive, and narrative-enhanced visualizations have proliferated in the last decade [1][7]. Several broad conclusions emerge regarding the Experimental Evidence on comprehension, recall, and engagement. Studies consistently show that narrative structures (sequencing, annotation) increase comprehension and recall for general audiences [11][23]. Author-driven stories often yield higher task accuracy when users must extract pre-defined insights [7][18]. User studies also indicate that interactivity sustains engagement and exploratory behavior [12][44]. However, unguided interactivity can increase cognitive load and cause users to miss important findings [5]. Empirical comparisons often find Balanced Configurations (hybrid designs) to be optimal, achieving both high engagement and comprehension, and often outperforming purely interactive designs [7][40].

The relative effectiveness of storytelling versus interactivity depends strongly on the Task Type as Moderator [4]. Fact-finding tasks (e.g., "What was the average delay?") benefit most from narrative guidance [4][7]. Conversely, exploratory tasks (e.g., "Identify unusual patterns") rely on interactive affordances [4][20]. For more complex sense-making and decision support tasks, hybrid designs that balance both are required [4][18]. Furthermore, Domain Expertise as Moderator shows that audience expertise also influences the preferred design. Studies

show that novices often prefer guided storytelling, while experts prefer open exploration with optional narrative scaffolding [25][31]. For mobility dashboards, this duality translates into designing for both commuters (novices) and analysts (experts)[8][9].

In order to explain the design heuristics and guidelines from literature, recent surveys and practitioner studies propose domain-specific heuristics for narrative visualizations and dashboards, which serve as a foundational guide for design and evaluation.

General visualization heuristics guide fundamental visual design principles:

- **Clarity over Complexity:** Minimize visual clutter by prioritizing legibility and focusing on relevant variables [7][31].
- **Consistency:** Use uniform encodings and interaction metaphors to reduce cognitive load [32].
- **Feedback and Affordance Visibility:** Ensure interactive controls are discoverable and provide immediate feedback [32].

While Narrative-Specific heuristics focus on how the story is constructed and consumed:

- **Annotation as Scaffolding:** Provide contextual annotations, but allow users to hide or reveal them for progressive disclosure [1][18].
- **Framing Transparency:** Clearly signal data sources, aggregation methods, and uncertainties to avoid rhetorical bias [6].
- **Progressive Disclosure:** Sequence insights from simple to complex, with options to dive deeper [20][39].

Moreover, the Dashboard-Specific heuristics (Mobility Context) address the unique constraints of spatio-temporal data and operational needs:

- **Spatio-Temporal Responsiveness:** Enable rapid filtering and manipulation across time and geography [2][41].
- **Cross-Device Adaptability:** Ensure that visualizations adapt seamlessly to mobile contexts without information overload [8][9].
- **Performance Optimization:** Dashboards must handle streaming or near real-time updates without lag, as delays can undermine usability [15].
- **Trust and Transparency:** For public-facing dashboards, explain methods of calculation (e.g., “average delay = mean minutes per trip”) to build public trust[6].

2.5 TRENDS AND EMERGING DEBATES IN DATA STORYTELLING

Beyond established heuristics, the literature points to several emerging trends and ongoing debates that shape the future of narrative visualization, particularly in complex domains like mobility.

Scrolly-telling and Immersive Formats are increasingly common in data journalism and offer a powerful way to guide the user's attention [23][39]. Studies find that integrating narrative progression with scrolling can improve attention and retention by providing a structured, linear flow that systematically guides the user's focus. This mechanism is effective for providing a clear narrative entry point. However, this format creates a tension with free exploration, as it may restrict the user's ability to investigate data laterally unless explicitly paired with branching interaction models or "stepper" interfaces that allow the user to break away from the author's path [39]. For mobility dashboards, finding responsive interaction patterns that successfully combine the guidance of scrolling with the detail of on-demand exploration remains a key design challenge.

The use of Automation and AI Assistance, such as large language model (LLM)-generated narratives and automated chart annotations, can significantly reduce the authoring effort required to produce complex data stories [19][38]. This acceleration in content creation holds immense promise for high-velocity environments where human analysis cannot keep pace with data volume. However, the literature warns against over-reliance on these systems: generated narratives may be inaccurate, hallucinate correlations, or present misleading summaries. Human oversight is necessary, particularly in sensitive contexts like transport policy, to ensure the veracity, contextual accuracy, and ethical framing of the generated story [19].

A recurrent and critical debate concerns the Ethical Concerns: Persuasion vs. Explanation regarding narrative visualization, specifically whether the act of storytelling risks manipulating audiences by emphasizing certain interpretations over others. Some scholars argue that storytelling is inherently persuasive and thus potentially manipulative, as the author chooses what to include and what to omit [22]. Conversely, others suggest that these risks can be significantly mitigated through transparent design choices, such as explicit labeling, clear data provenance, and communicating uncertainty [6]. For mobility operators, balancing persuasive communication (aimed at building public trust and highlighting service improvements) with transparent reporting (essential for accountability and user sense-making) is a critical design and policy challenge.[22]

2.6 RESEARCH GAP AND STUDY CONTRIBUTIONS

The reviewed literature establishes that mobility data visualization is characterized by four intersecting challenges: balancing narrative guidance with exploration [4][39], managing spatio-temporal complexity [2][17][41] navigating evaluation diversity [13][40], and addressing trust and ethical framing [6][22]. Despite these foundational insights, current practice and research suffer from specific deficiencies that this study addresses.

Current public-facing transport dashboards, often derived from legacy systems used for operational monitoring, generally fail to address the specific informational needs of commuters [18].

- **Static and Linear Constraints:** Existing systems present fixed, pre-calculated aggregates (e.g., "average delay") in linear formats. This lack of user agency prevents filtering and isolation of personally relevant data, compromising engagement [36] and failing to convey the dynamic, temporal evolution of mobility patterns [2].
- **Cognitive Overload and Design Complexity:** When interactivity is included, it often attempts to display all metrics simultaneously. This high information density violates the principle of progressive disclosure, leading to visual clutter and high cognitive load [32].
- **Mobile Incompatibility:** Many complex, desktop-centric layouts are not optimized for mobile devices. Given that commuters predominantly access information via smartphones while in transit, this incompatibility renders the tools practically useless in real-world scenarios [8][9][41].
- **Lack of Narrative Scaffolding:** Legacy dashboards function as data repositories that describe *what* is happening but fail to provide the narrative context (annotations, sequencing) to explain *why* it matters, thus restricting sense-making for non-expert users.
- **Mobile Usability Shortfall:** Few studies address cross-device storytelling in mobility contexts [2]. Research is needed to validate narrative patterns optimized for small screens in this domain.[41]

Synthesizing the limitations of current systems with the theoretical literature reveals specific, actionable deficiencies that this study is designed to address. The overarching research problem stems from a scarcity of domain-specific research at the intersection of information visualization and narrative visualization, particularly within the complex, high-velocity spatio-temporal domain of mobility [1][2][41].

Specifically, this study is designed to overcome three primary gaps in current literature and practice:

1. **Empirical Hybrid Comparison Gap:** The literature calls for in-situ studies under realistic constraints [2][41], yet there is a lack of rigorous, domain-specific research that empirically tests a hybrid model combining strong narrative scaffolding with user-driven interactivity within the public transport sector.
2. **Lack of Standardized Protocol:** While heuristic sets exist for narrative visualization [13] and for dashboards [18] separately, a validated, integrated evaluation instrument tailored to mobilityone that combines usability, narrative clarity, and efficiency (TTI)is scarce [1][7].
3. **Rhetorical Metric Gap:** Methods for empirically evaluating how narrative choices (framing, color, aggregation) in public transport dashboards influence rider perceptions of trust, fairness, and service quality are underdeveloped [22].

This study directly addresses these deficiencies by creating and rigorously testing a hybrid visualization approach. The core theoretical premise is that a system synthesizing the **explanatory** power of data stories with the **exploratory** power of interactive dashboards, specifically tailored for the complexities of the Deutsche Bahn mobility ecosystem, will yield optimal usability.

The key contributions of this research are therefore defined as:

- **Empirical Validation in Commuter Contexts:** Providing the domain-specific evidence requested by the literature [1][41] through the evaluation of commuter-facing prototypes using real Deutsche Bahn data.
- **Standardized Protocol Development:** Proposing a reusable, integrated protocol that combines measures for usability i.e., UEQ, narrative clarity, and TTI, setting a new standard for hybrid system assessment [1][7][18].
- **Mobile Usability Insight:** Generating novel empirical insights into cross-device storytelling and optimal interaction patterns for complex, narrative-rich dashboards on mobile platforms.
- **Rhetorical Analysis:** Contributing to the ethical dimensions of public-facing transport visualization by analyzing how the dashboard's design influences user perception of trust and fairness in mobility data [6][22][23].

FOCUS GROUP

This chapter presents the preparation, implementation, and evaluation of two focus group sessions conducted as part of the design phase of a mobility-focused data storytelling dashboard. Since dashboards represent a combination of visual analytics and narrative elements, understanding how users interpret, interact with, and emotionally perceive dashboard information is crucial. A focus group supports the development of a dashboard containing storytelling elements as it provides early, exploratory insights into how potential users understand the information, what they pay attention to, and what they find confusing or meaningful. Unlike other methods, such as surveys, focus groups allow participants to talk together, which reveals mental models, expectations, and values. These insights are fundamental for designing a dashboard that not only shows the data and visualizations but tells a compelling, intuitive story. The focus groups provided early qualitative insights that informed the conceptual design of the prototype developed later in the project.

3.1 PREPARATION

The preparation stage played a significant role in ensuring that both focus groups sessions produced reliable, comparable, and meaningful data. Because the goal was to explore user expectations, visual preferences, and interpretative patterns, careful planning was required for the selection of participants, choice of methods, formulation of questions, and organization of materials.

The overall intention of the preparation phase was to create an environment where participants could comfortably share their experiences and perspectives, while also being guided through a structured exploration of two existing public dashboards. This structure allowed the research team to gather insights about narrative flow, clarity, data visualization quality and interactive elements, which are essential components of good dashboard storytelling.

3.1.1 *Goals and Methods*

The primary goal of both focus groups was to collect qualitative user insights for shaping the early conceptual design of a mobility dashboard prototype. The research emphasized storytelling-oriented dashboard design, so qualitative methods were more suitable than quantitative survey approaches.

The focus group method was selected because it encourages:

- spontaneous feedback from multiple participants,

- discussions that reveal reasoning behind preferences,
- discovery of pain points through shared experiences,
- contrastive evaluation when multiple dashboards are presented,
- co-creation through whiteboard sketching.

Thinking-aloud: The thinking-aloud method was used while participants explored the dashboards. They were encouraged to verbalize their thoughts in real time to what they noticed, what confused them, and how they interpreted the visuals. This method helps uncover users' immediate reactions, cognitive processes, and interpretation challenges that are not visible through observation alone.

Idea cards: Idea cards were provided after the exploration phase to capture participants' suggestions in a structured yet creative format. They were asked to write their observations, improvements, missing elements, or new features that they would find to be valuable. This method supports divergent thinking and ensures all participants' ideas are collected. Idea cards helped identify storytelling elements such as annotations, context, and features that users felt were missing.

Sketching: In the final phase of the focus group, participants were asked to use the whiteboard to sketch their ideal Dashboard based on what they observed earlier. Sketching encourages participants to express their thoughts visually, making it easier to communicate layout preferences, navigation flows, or storytelling structures. This method helped translate abstract ideas into concrete design directions for the prototype.

Two dashboards were chosen to stimulate discussion:

- Belgian Mobility Dashboard [3]
- Klima Dashboard [24]

We selected these dashboards based on a set of practical and qualitative criteria that supported our research goal of understanding how a mobility dashboard could be transformed into a data story.

Relevance: The Belgian Mobility dashboard offered the most comprehensive and detailed view of the mobility data among the options we reviewed. It covered multiple domains (e.g., traffic, public transport, EVs, heatmaps) and had a strong multimodal perspective, which made it a suitable benchmark for exploring how complex mobility information can be communicated effectively.

Strength: As it presented rich, layered insights, the mobility Dashboard provided a good foundation for analyzing where storytelling elements were missing and what kind of narrative guidance users might need.

UX/UI Design: The Klima Dashboard was chosen as we viewed it as a good example of UX/UI Design, including clear visual hierarchy, strong aesthetics, and features such as dark mode. While it was less mobility-specific, it provided a good example of user-friendly design

that could inspire the prototype.

Alternatives: We also reviewed other dashboards, including regional mobility dashboards from other European regions. However, we felt that they either lacked the required detail or did not demonstrate the design qualities we aimed to evaluate. For these reasons, we ultimately focused on the two dashboards that best aligned with our research objectives.

Belgian Mobility Dashboard [3]: The Belgian Mobility Dashboard is a comprehensive, data-rich platform that provides an overview of mobility patterns across Belgium. Visually, the dashboard uses a clean, light interface with a modular layout consisting of multiple panels and charts. It includes interactive filters (e.g., time range, region, transport mode), map visualisations, line charts, bar charts, and key indicators summarising traffic flow, congestion levels, travel times, public transport activity, and cycling trends. One of its strengths is the multimodal structure, allowing users to switch between different transport categories such as road traffic, rail, cycling, and freight. It also integrates real-time or regularly updated data, gives access to historical trends, and provides tooltips that reveal additional details.

Klima Dashboard [24]: The Klima Dashboard presents environmental and climate-related indicators with a strong emphasis on visual clarity and modern design. It features a dark-mode interface, high-contrast color palettes, and a visually appealing layout with large, easy-to-read charts. The dashboard includes interactive graphs on emissions, energy consumption, renewable energy production, and climate targets. Each section is structured around a thematic category, and charts often include simple annotations or reference values. Although the content is not mobility-specific, the interface demonstrates effective UX/UI practices, such as minimal visual clutter, consistent iconography, and smooth navigation between pages. Its design illustrates how aesthetics and structure can improve comprehension and engagement valuable inspiration for a storytelling-oriented prototype.

3.1.2 Formulation of the Interview Questions and Procedure

The interview questions were designed to follow a progressive structure that helps participants gradually open up and move from simple reflections to detailed analysis. The main considerations behind the design of the questions were:

1. **Reducing participant pressure.** The session began with simple open questions such as: – “Have you used dashboards before?” – “What is your understanding of a dashboard?” These questions establish a baseline of familiarity and help participants relax.
2. **Establishing context.** After warming up, participants were invited to explore the first dashboard. They were encouraged to take their time, scroll freely and share their initial impressions.

3. **Guided evaluation.** After initial exploration, the moderator introduced guided prompts:

- “What are your immediate thoughts on this dashboard?”
- “What makes a dashboard helpful or frustrating for you?”
- “What storytelling elements did you find helpful here (e.g., timelines, annotations, highlights)?”
- “How does this dashboard compare to the others in terms of telling a story?”
- “What’s usually missing or hard to understand in dashboards you encounter?”

These prompts focused attention on narrative flow, data clarity and visual hierarchy.

4. **Comparative questions.** Once both dashboards had been explored, participants were asked to compare the two. Comparative evaluation produces clearer and more detailed opinions because differences become more apparent.

5. **Creative closing activity.** Participants designed their own ideal dashboard on a whiteboard. This helped reveal the features they value most, and the structural mental model they have when imagining a “good” dashboard.

The procedure was explained beforehand, and participants were informed that the sessions would be recorded for analysis. They were also reassured that there were no right or wrong answers and that their task was simply to share their experiences and thoughts.

3.1.3 *Selection of Participants and Environment*

Two focus groups were planned:

- **Focus Group A** – 7 participants
- **Focus Group B** – 7 participants

For the focus groups, we aimed to recruit a diverse sample of participants with different academic, professional, and personal backgrounds. Our goal was to include individuals with varying levels of familiarity with mobility data, dashboards, and data visualisation in general. This diversity would have enabled us to capture a broader range of perspectives and reduce bias in how the dashboards were interpreted. Ideally, the group would have included commuters, students, working professionals, and people with experience in data analysis or mobility planning allowing us to compare how different users understand and engage with mobility data stories.

The usability laboratory environment at TU Ilmenau was selected for the sessions to ensure:

- controlled acoustics and lighting,
- uninterrupted discussion conditions,
- availability of large displays,
- recording equipment for data capture,
- space for individual and group sketching.

The environment supported both comfort and focus, enabling deeper engagement with the dashboard content.

3.1.4 *Creating the Focus Group Documents*

To ensure consistency and structure across both focus groups, several documents and materials were prepared:

- **Moderator script:** ensuring identical introduction and session flow.
- **Open question sheets:** for warm-up and initial reflections.
- **Guided question cards:** to prompt discussion on usability, storytelling and clarity.
- **Idea mapping cards:** allowing participants to categorize positive and negative aspects.
- **Whiteboard templates:** assisting participants in sketching their ideal dashboard layout.

These documents ensured that the focus groups were comparable in structure and quality, reducing variability introduced by the moderator.

3.2 IMPLEMENTATION OF THE FOCUS GROUPS

Both sessions followed the same structure and timeline to ensure comparability. The moderator guided the sessions but encouraged participants to respond freely and interact with one another. Each focus group lasted approximately 90 minutes. The first focus group had 5 participants, whereas the second group had 6. The participants were students of TU Ilmenau. Although this is not a representative sample of the general population, it is appropriate for the early stage of conceptual design, where the goal is to collect informed feedback rather than statistically generalizable data.

The structure of both groups was:

1. Welcome and warm-up
2. Exploration of Dashboard 1 [3]

3. Exploration of Dashboard 2 [24]
4. Group comparison and discussion
5. Whiteboard sketching and closing remarks

Order of Dashboards: To account for potential order effects, the sequence in which the dashboards were presented differed between the two focus groups. In Focus Group 1, participants first explored the Klima Dashboard [24] before moving on to the Belgian Mobility Dashboard [3]. In Focus Group 2, this order was reversed: participants began with the Belgian Mobility Dashboard [3] and subsequently examined the Klima Dashboard [24].

This counterbalanced approach ensured that participants' evaluations were not unintentionally influenced by the position of the dashboard within the session. By alternating the order, we aimed to minimise biases such as increased familiarity, fatigue, or the tendency to compare the second dashboard more critically. This allowed us to better isolate feedback related to the dashboards' content and design, rather than the sequence in which they were viewed.

It is to be noted that we switched the order of dashboards in the second focus group to see how it impacts the results. Participants were encouraged to verbalize their thoughts ("think aloud") during the exploration. This method helped uncover initial impressions and cognitive processes related to navigation, visual search, and interpretation.

3.3 EVALUATION OF THE FOCUS GROUPS

The evaluation consists of structured session summaries and detailed qualitative analysis. Each focus group is presented in chronological order, followed by an interpretation.

Time	Task	Method	Key Findings
14:09	Welcome warm-up	+ Open questions	Participants shared expectations and experience with dashboards.
14:22	Explore Dashboard 1	Exploration + idea cards A.5	Clear, compact, strong storytelling; missing timeline, search bar, dark mode.
14:56	Explore Dashboard 2	Exploration + idea cards A.5	Attractive but cluttered; navigation issues; overloaded with information.
15:07	Compare dashboards	Group discussion	Preference for Belgian Mobility Dashboard (5/5).
15:23	Design wrap-up	+ Reflection + sketching 3.1	Common issues: lack of detail in charts, slow loading, need for better explanations.

Table 3.1: Analysis of Focus Group A

Focus Group A consistently prioritized clarity, simplicity and strong visual hierarchy. The Belgian Mobility Dashboard [3] was immediately perceived as easier to navigate due to its clean structure, balanced use of white space and linear storytelling approach.

Participants appreciated:

- the compact layout,
- the structured progression of information,
- the clear grouping of content,
- the absence of overwhelming visual elements.

The Klima dashboard [24] generated mixed reactions. Although visually appealing, participants described it as:

- too dense,
- lacking a clear focal point,
- difficult to navigate,
- overwhelming for first-time users.

The unanimous preference for the Belgian dashboard [3] highlights the importance of explicit visual storytelling in dashboards that aim to inform broad audiences.

Time	Task	Method	Key Findings
16:14	Welcome warm-up	+ Open questions	Participants' expectations and dashboard familiarity collected.
16:25	Explore Dashboard 1	Exploration + idea cards A.6	Good design and interactivity; cluttered visuals; poor mobile usability.
17:02	Explore Dashboard 2	Exploration + idea cards A.6	Clear and compact; missing advanced features (timeline, search, dark mode).
17:17	Compare dashboards	Group discussion	Strong preference for Klima Dashboard (6/6).
17:35	Design wrap-up	+ Reflection + sketching 3.2	Issues: slow loading, insufficient chart explanations.

Table 3.2: Analysis of Focus Group B

Unlike Group A, this group favored the Klima Dashboard [24] due to:

- perceived flexibility,
- better navigation depth,
- interactive customization elements,
- more modern visual design.

Participants in this group appeared more experienced with data-heavy systems and valued dashboards that allow deeper exploration. However, they still identified important issues such as clutter, lack of clarity in some charts and insufficient contextual explanations.

Both groups independently identified similar issues across both dashboards [3] [24]:

- slow loading times,
- insufficient explanations for charts,
- need for clearer narrative structure.

Interpretation of the Tables and Summary of Participant Responses:

The tables 3.1 and 3.2 provide a structured overview of the sequence and duration of the two focus group sessions. Both focus groups lasted approximately 90 minutes, progressing through a warm-up phase, the exploration of two dashboards, a comparative discussion, and a final sketching exercise. While the tables outline the procedural flow, the following sections summarise the participants' main responses and impressions for each task.

Warm-up: Expectations and prior experience: During the warm-up, participants were invited to share their expectations and previous experience with dashboards. Across both groups, participants reported having used dashboards primarily in academic contexts or for general data access (e.g., statistics portals, study projects). They expressed that dashboards often appear “too technical” or overloaded, and they expected a mobility dashboard to present trends clearly, avoid unnecessary complexity, and provide accessible explanations of unfamiliar indicators. Several participants emphasised the importance of intuitive navigation and a clearer narrative that helps them understand why specific metrics matter.

Exploration of Dashboard 1: The first dashboard (the Klima Dashboard [24] in Focus Group A and the Belgian Mobility Dashboard [3] in Focus Group B) elicited immediate comments through the thinking-aloud method.

Klima Dashboard [24]: Participants consistently praised its visually appealing dark-mode design, clean layout, and accessible charts. They found the interface modern and “pleasant to look at.” However, they also remarked that some of the explanations accompanying charts appeared brief, leaving them wondering about definitions or calculation methods.

Belgian Mobility Dashboard [3]: Participants appreciated the depth and breadth of the mobility indicators, noting that the multimodal perspective (traffic, public transport, cycling, freight) offered a detailed picture of mobility in Belgium. At the same time, they found the interface information-dense and occasionally difficult to navigate. Several noted that some graphs lacked contextual cues, which made interpretation more challenging and reduced the perceived narrative clarity.

Exploration of Dashboard 2: For the second dashboard, participants naturally compared their impressions to the first one they viewed. When the Mobility Dashboard [3] followed the Klima Dashboard [24], it was perceived as far more detailed but also significantly more overwhelming. Participants felt they needed more guidance to understand the relationships between indicators. When the Klima Dashboard [24] appeared second, participants described it as “visually calming” and easier to process after the more complex Mobility Dashboard [3]. This contrast confirmed that the dashboard sequence influenced user impressions a key reason why the order was counterbalanced across the two groups.

Group comparison and discussion: During the comparative discussion, participants in both focus groups expressed a clear preference for the Belgian Mobility Dashboard [3], mainly because it provided more substantive mobility information. Nevertheless, they criticised its limited contextualisation and lack of storytelling elements. In contrast, the Klima Dashboard [24] was viewed as aesthetically superior, more readable, and easier to navigate, yet less informative for mobility-specific purposes. Participants concluded that the ideal dashboard would combine the rich, multimodal content of the Mobility Dashboard [3] with the clarity, structure, and visual appeal of the Klima Dashboard [24].

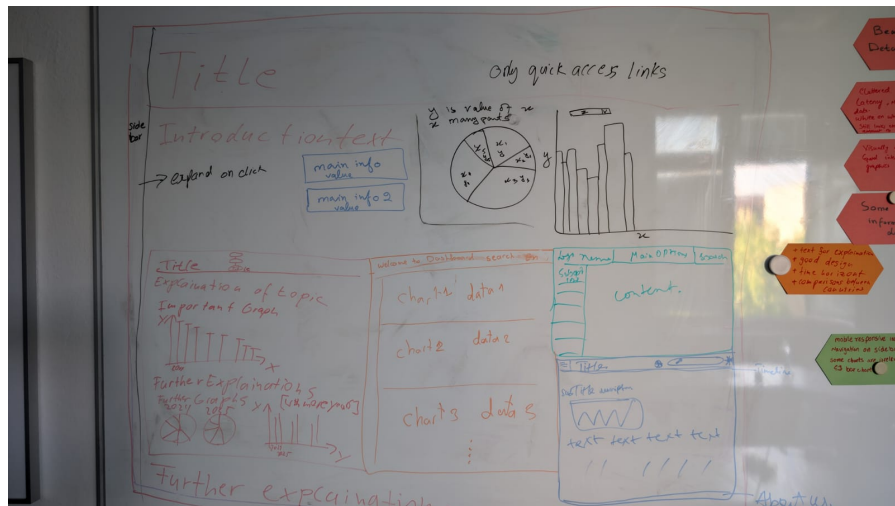


Figure 3.1: Sketching of Dashboards by Participants for the first Focus Group session.



Figure 3.2: Sketching of Dashboards by Participants for the second Focus Group session.

Sketching and closing reflections: In the final sketching activity, as shown in 3.1 and 3.2, participants visualised improvements they would like to see in a future prototype. Common suggestions included integrating brief narrative explanations next to charts, adding a timeline slider or comparison function, reducing visual clutter, grouping related indicators more clearly, and offering a dark-mode option. In their closing reflections, participants emphasised that a mobility dashboard becomes more understandable when it highlights key insights, provides contextual guidance, and reduces unnecessary complexity. They noted that such improvements would not only support comprehension but also increase their motivation to explore the data further.

The qualitative analysis integrates insights from both groups and identifies key themes that emerged repeatedly throughout the discussions.

Importance of Storytelling Flow: Participants emphasized the importance of a dashboard that guides the viewer through data in a logical order. Dashboards with a strong narrative structure where visuals are arranged to support sequential understanding, were consistently rated as more intuitive and pleasant to use.

Need for Interactivity: Interactive elements such as filters, timeline sliders, search functions, and dark/light mode were described as essential. These elements help users explore data at their own pace and create personalized views that support individual interpretation.

Visual Clarity and Minimalism: Both groups expressed a preference for dashboards with:

- restrained color palettes,
- clean iconography,
- sufficient spacing,
- clear labels and legends.

Minimalistic design was associated with reduced cognitive load and improved storytelling.

Explanatory Context: Participants noted that dashboards require explanatory elements such as legends, annotations and descriptions to help users interpret data correctly. The absence of such elements forced users to guess or infer meaning, reducing trust and making interpretation less efficient.

Performance and Technical Issues: Both dashboards [3] [24] were reported to have loading delays. Participants described such delays as disruptive and stated that they interrupt the natural flow of exploration. Smooth performance was perceived as part of the overall experience and critical for maintaining user engagement.

PROTOTYPE AND IMPLEMENTATION OF THE DASHBOARD

4.1 USER FLOW AND NARRATIVE STRUCTURE

The application is designed with a linear narrative entry that expands into an exploratory dashboard. The user flow consists of three distinct phases:

Phase 1: The Entry Point (Homepage)

The landing page serves as the introduction. It features a horizontal navigation bar with three primary options: **Home**, **Interactive Map**, and **Contact**. Users are free to explore these tabs without restriction.

However, the primary call-to-action is a prominent button at the bottom-centre of the screen labelled as, Start Interactive Quiz. This button serves as the narrative hook. By pressing it, the user initiates the guided storytelling experience, transitioning from a passive observer to an active participant.

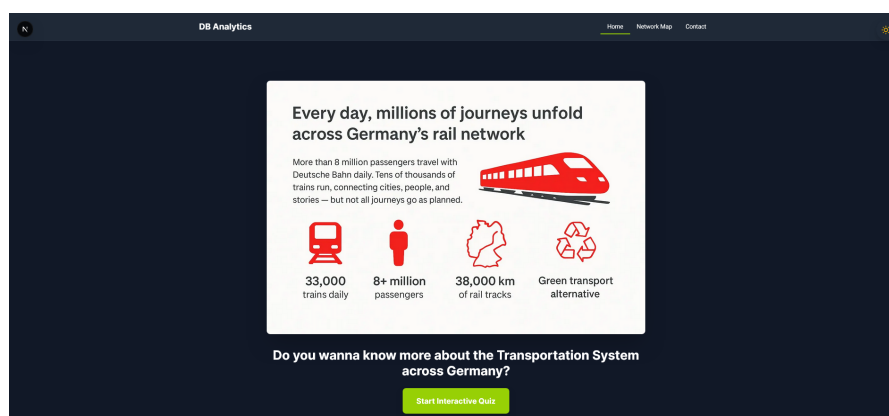


Figure 4.1: Dashboard - The Homepage [27]

Phase 2: The Interactive Quiz (The Hook)

Once the quiz begins, the user is presented with questions based on recent facts and trends regarding mobility in Germany. The logic follows a Challenge-Feedback loop:

1. **Challenge:** The user selects an answer for the current question (Correct or Wrong).
2. **Immediate Feedback:** The system validates the answer, displaying whether it is true or false.
3. **Contextual Data:** Crucially, after the answer is revealed, the system displays relevant charts or figures referencing that particular quiz question, along with a descriptive text. It ensures that users leave the screen with the correct data point and context.

4. **Progression:** The user advances to the next question via an arrow button in the bottom-right corner.

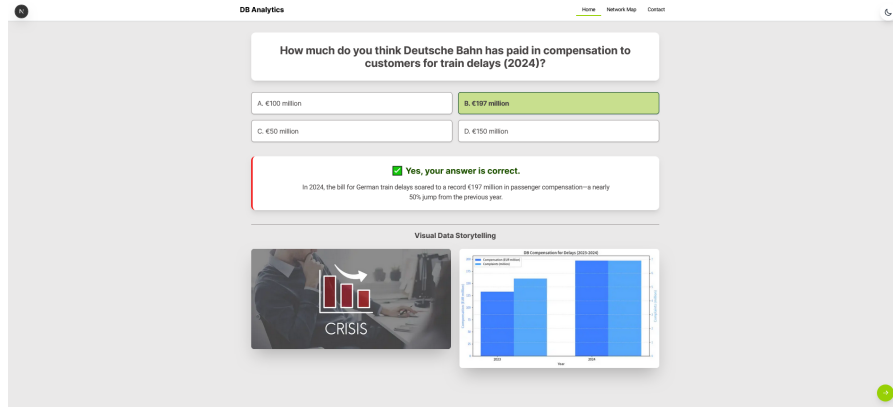


Figure 4.2: Dashboard - The Interactive Quiz [27]

This linear flow continues for three questions. On the final quiz question, the arrow button is replaced by a “VIEW DASHBOARD” button.

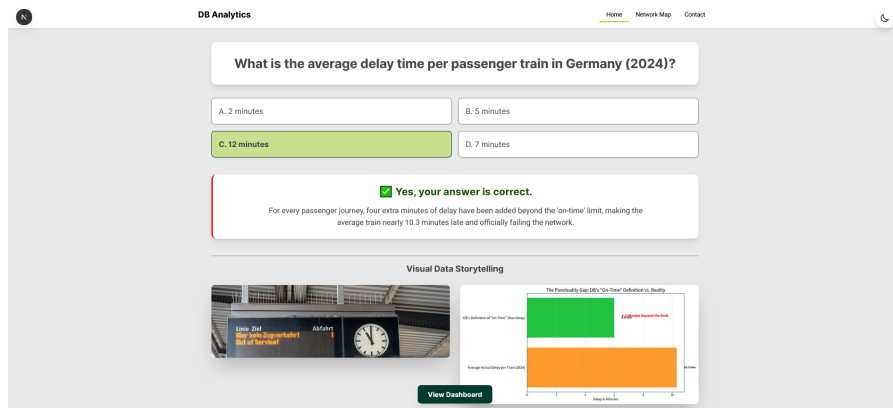


Figure 4.3: Dashboard - VIEW DASHBOARD [27]

Phase 3: The Dashboard (Exploration)

Clicking “VIEW DASHBOARD” transitions the user to the primary data interface. Here, the structure changes from linear to exploratory. A vertical sidebar presents five distinct metrics (Punctuality, Congestion, Incidents, Multi-modal, Environment). Users can navigate these freely to view detailed graphs and figures supporting the storytelling.

Note on Navigation Constraint: While users can switch between the vertical tabs freely, the design intentionally prevents a direct return to the quiz start screen without reloading. If the user navigates to another horizontal tab (e.g., Contact), they must restart the quiz to re-enter the flow, enforcing the narrative progression from “Quiz” (Learning) to “Dashboard” (Analysing).

4.2 STRUCTURE AND CONTENT OF THE QUIZ

The quiz functions as a preparatory layer, priming the user with high-level facts before they dive into granular data. It consists of three multiple-choice questions derived from 2024 mobility reports.

The following questions and answers are implemented in the final prototype:

- **Question 1:** “How much did DB pay in compensation for delays in 2024?” [A.4](#)
 - *Correct Answer:* €197 million
 - *Key Detail/Context:* €197 million represents a **50% increase** from the previous year.
 - *Visual Asset:* A chart comparing compensation costs year-over-year.
- **Question 2:** “What is the most common cause of train delays?” [A.2](#)
 - *Possible Answer:* Infrastructure bottlenecks
 - *Key Detail/Context:* Infrastructure bottlenecks account for **80%** of all train delays.
 - *Visual Asset:* A breakdown graph showing causes of delays.
- **Question 3:** “What’s the average delay time per train?” [A.3](#)
 - *Correct Answer:* 10.3 minutes
 - *Key Detail/Context:* 10.3 minutes is officially recognised as failing the network’s minimum standard.
 - *Visual Asset:* A timeline graphic illustrating average delay trends.

Each question includes an explanation and visual assets that are displayed immediately after the user submits their answer.

4.2.1 Rationale for the Interactive Quiz

The decision to utilise an interactive quiz for specific user needs and narrative purposes drove the primary storytelling element:

1. **Active vs. Passive Engagement:** Standard dashboards often present data passively. By forcing the user to make a choice (e.g., guessing an answer), the quiz creates an “information gap”. Users become curious to see if they are correct, increasing their attention span for the subsequent data reveal.
2. **Correcting Misconceptions:** The quiz uses empirical data to ground the user in facts (e.g., the massive cost of compensation) before they look at the complex charts.

3. **Gamification:** The quiz format introduces a ludic element to a serious topic. It lowers the barrier to entry for non-expert users who might otherwise feel overwhelmed by a complex dashboard immediately upon arrival.

4.3 JUSTIFICATION OF DATA SELECTION

The data sets selected for both the quiz and the main dashboard (Punctuality, Congestion, Incidents, Multi-modal, Environment) were chosen to provide a holistic view of the current Mobility Crisis in Germany [35] [45].

- **Relevance to Public Interest:** Punctuality and Compensation (Quiz Q1 & Q3) were chosen because they are the most immediate pain points for daily commuters [26].
- **Root Cause Analysis:** Infrastructure data (Quiz Q2) was selected to explain *why* the delays happen, moving the narrative from symptoms to causes.
- **Connection to Dashboard:** The quiz data serves as the “headline”, while the dashboard data serves as the “article”. For example, the quiz states that delays cost €197 million; the dashboard then allows users to explore specific punctuality trends that contributed to that cost [26].

This structured approach ensures that the user not only sees numbers but also understands the story behind the mobility network’s current state.

4.4 CONCEPTUALISATION AND INSPIRATION

The practical component of this project began with identifying a significant gap in the German media landscape: the lack of a centralized, user-friendly and narrative-driven data dashboard for national mobility. Although raw data exists [16] [35], they are often disparate and not presented in a context that facilitates public understanding.

The primary inspiration for this endeavor was the Belgian Mobility Dashboard [3]. An initial analysis of this dashboard revealed an effective combination of data visualization, interactivity, and a clear user interface, which served as a benchmark for the project's own design goals.

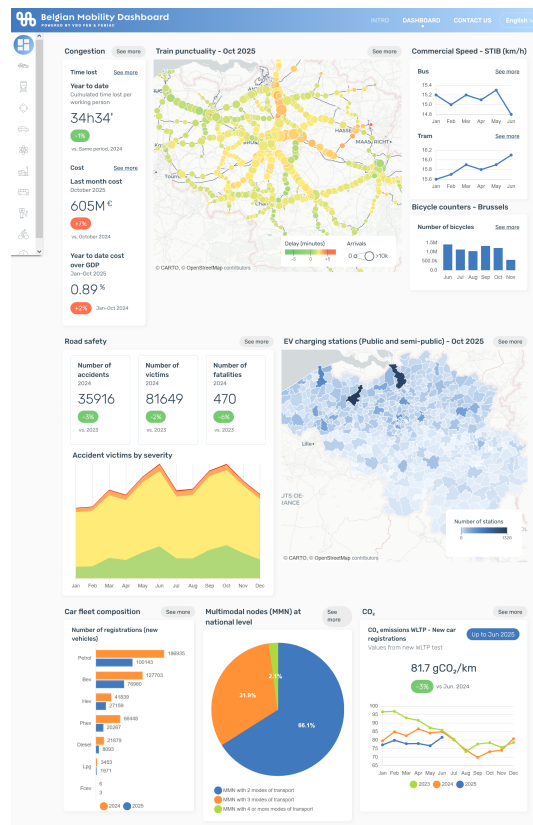


Figure 4.4: The Belgian Mobility Dashboard. The dashboard visualizes key 2025 transport metrics, ranging from economic indicators like congestion costs to social factors like road safety and train punctuality. It also tracks the environmental transition, displaying current data on EV charging infrastructure. [3]

To further refine the concept and validate these initial observations, a Focus Group Discussion (FGD) was conducted between the end of July and the beginning of August. During this session, participants were presented with two exemplary dashboards and asked to provide critical feedback. The discussion focused on defining what an effective dashboard should look like, specifically regarding User Interface (UI) design, data storytelling elements, and the presentation of charts and

graphs. The participants highlighted the importance of clear insights, noting that a successful website must guide the user through the information. This feedback was instrumental in shaping the requirements for the project.

Based on inspiration from the Belgian model [3] and the insights from the FGD, the central concept was solidified: to develop a dashboard that not only present data but also guides the user through a narrative, combining quantitative facts with qualitative, story-driven context.

4.5 INITIAL DESIGN AND PROTOTYPING

Following the project's prescribed steps, the initial design phase was conducted using the UI/UX design tool, Figma [14]. Several wireframes [14] and low-fidelity mockups were created to explore potential layouts, user flows, and visual styles. This process was valuable for establishing a basic information architecture.

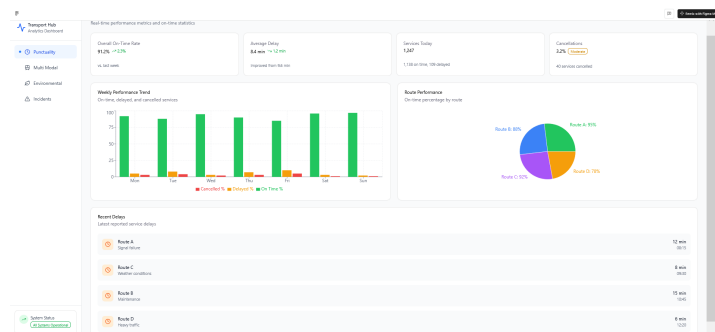


Figure 4.5: Figma - The Punctuality Overview Metric [14]

However, it became apparent that static Figma prototypes [14] were insufficient for adequately exploring the dynamic and highly interactive elements that were central to the project's hypothesis. The goal was not just to design a static layout, but to test the *usability of interactive storytelling*. Therefore, the decision was made to move to a high-fidelity prototype where these interactions could be built and tested directly.

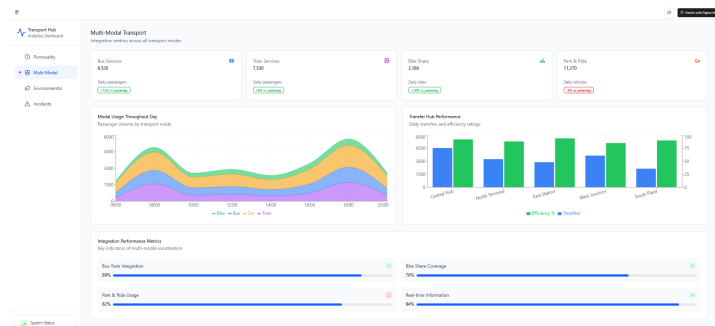


Figure 4.6: Figma - The Multi-Modal Metric [14]

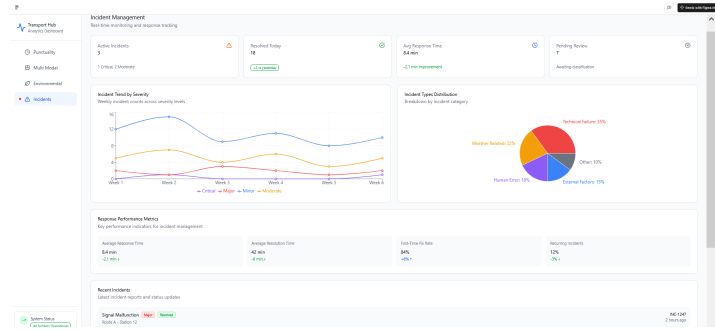


Figure 4.7: Figma - The Incidents Metric [14]

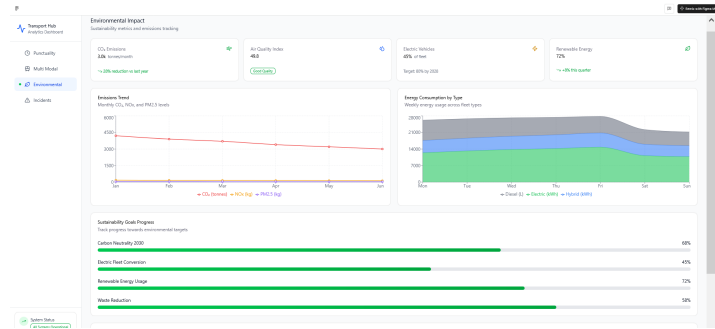


Figure 4.8: Figma - The Environmental Metric [14]

4.6 IMPLEMENTATION AND TECHNOLOGY STACK

The development of the dashboard prototype [27] began in mid-October and concluded when the final version was displayed at the beginning of November. Following this implementation phase, the final usability testing was conducted to evaluate the effectiveness of the design.

To build this high-fidelity prototype, the development was initiated using Visual Studio Code as the primary code editor. A modern, robust technology stack was selected to ensure performance, scalability, and a seamless user experience. The core technologies utilized in the implementation include:

- **Next.js & React:** The dashboard was built using Next.js, a React framework. React's component-based architecture allowed for the creation of reusable, stateful UI elements (such as dynamic charts and navigation), while Next.js provided the necessary structure for routing and optimized performance.
- **TypeScript:** To enhance code reliability and maintainability, TypeScript was employed. Its static typing system helped prevent common errors during development and ensured that data structures particularly those used for the diverse mobility metrics were handled consistently throughout the application.
- **Tailwind CSS:** For styling, the utility-first CSS framework Tailwind CSS was used. This allowed for rapid UI development and

ensured the dashboard was responsive and visually consistent across different screen sizes, a key factor for usability.

- **Leaflet & OpenStreetMap:** A central feature of the dashboard is its geospatial interactivity. [33] This was achieved using Leaflet, a leading open-source JavaScript library for mobile-friendly interactive maps. The map data itself was sourced from OpenStreetMap, providing the underlying geographic context required to visualize mobility data effectively across Germany.

This combination of tools allowed for rapid iteration, building upon the initial concepts from the Figma wireframes [14] to create a fully functional, really interactable application.

4.7 DASHBOARD ARCHITECTURE: STORYTELLING THROUGH DATA

The final dashboard [27] is a web-based application designed to tell a comprehensive story of mobility in Germany. The narrative is structured around five key thematic areas, each presented as a distinct, interactive tab. The five core “data stories” of the dashboard are:

1. **Punctuality:** Visualizing the performance and reliability of the Deutsche Bahn (DB) and other transport services.

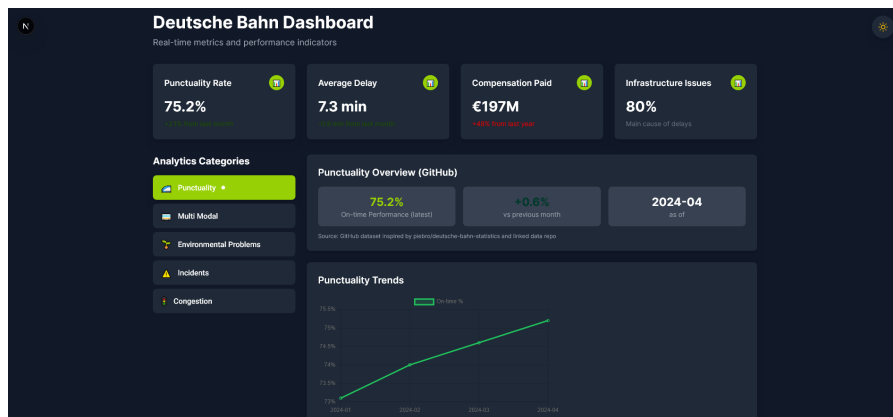


Figure 4.9: Dashboard - Punctuality Metric [27]

2. **Congestion:** Presenting data on road and rail network usage, peak times, and bottlenecks.

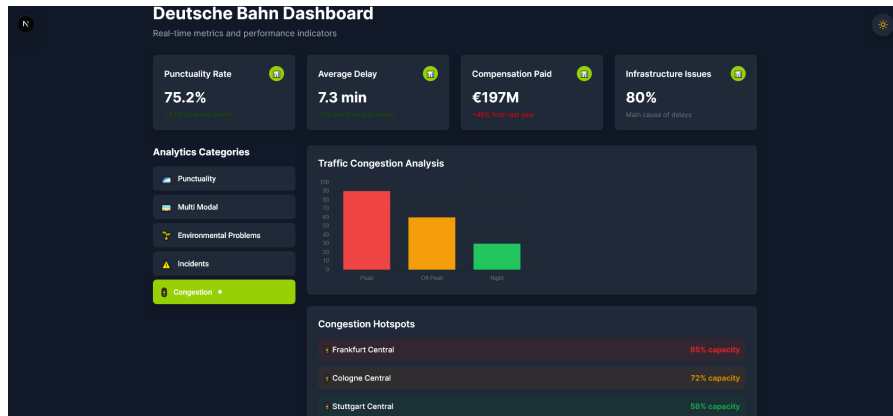


Figure 4.10: Dashboard - Congestion Metric [27]

3. **Incidents:** Mapping and explaining the impact of disruptions, construction, and accidents on the mobility network.

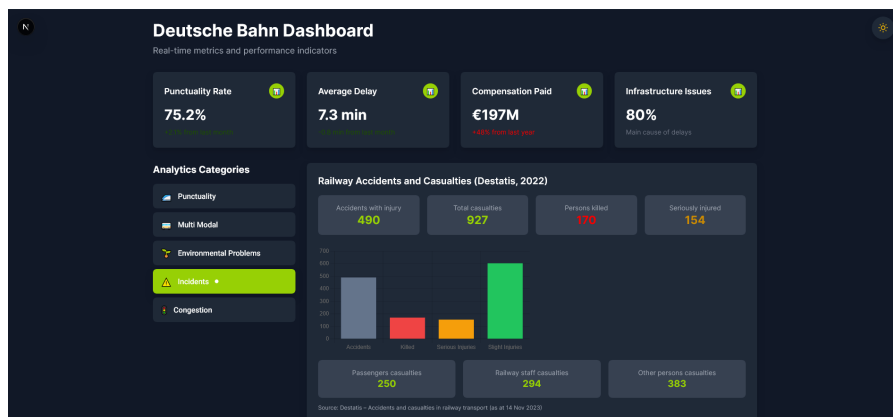


Figure 4.11: Dashboard - Incidents Metric [27]

4. **Multi-Modal Options:** Showcasing the availability and integration of different transport types (e.g., bike-sharing, public transit, car-sharing).

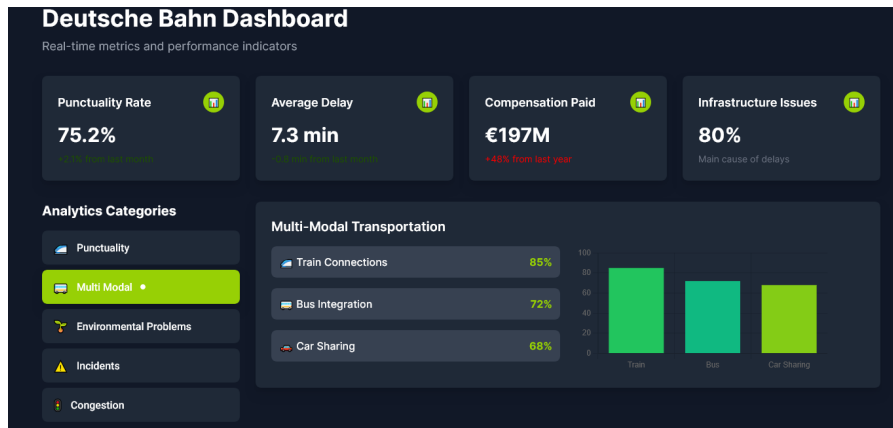


Figure 4.12: Dashboard - Multi-Modal Metric [27]

5. **Environmental Impact:** Telling the story of the environmental costs and benefits associated with Germany's mobility choices, including emissions and green initiatives.

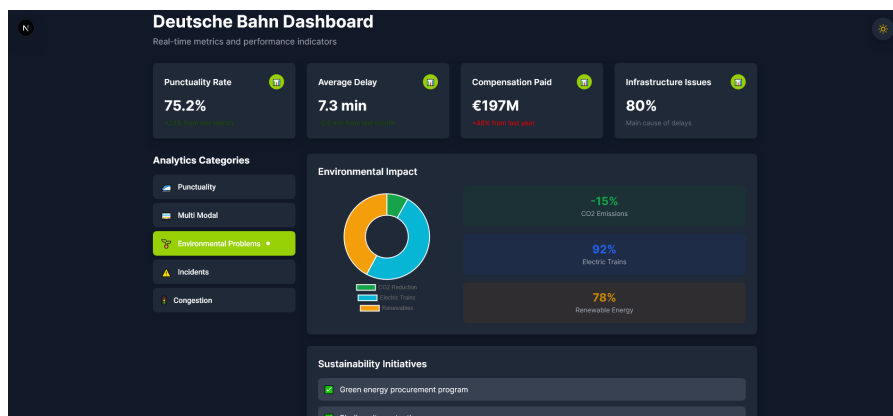


Figure 4.13: Dashboard - Environmental Metric[27]

To bring these stories to life and fulfill the project's requirement for interactivity, several key features were implemented:

- **Interactive Map:** A central map [27] enables users to explore geospatial data across all five themes, filter information, and observe regional patterns.

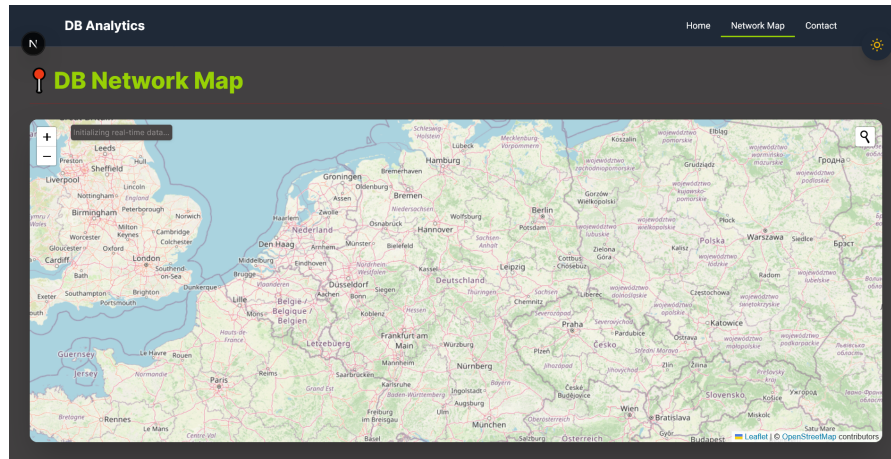


Figure 4.14: Dashboard - Interactive Map [27]

- **Interactive Quiz:** To enhance user engagement and knowledge retention, an interactive quiz tests [27] the user's understanding of the facts presented, reinforcing the data story.

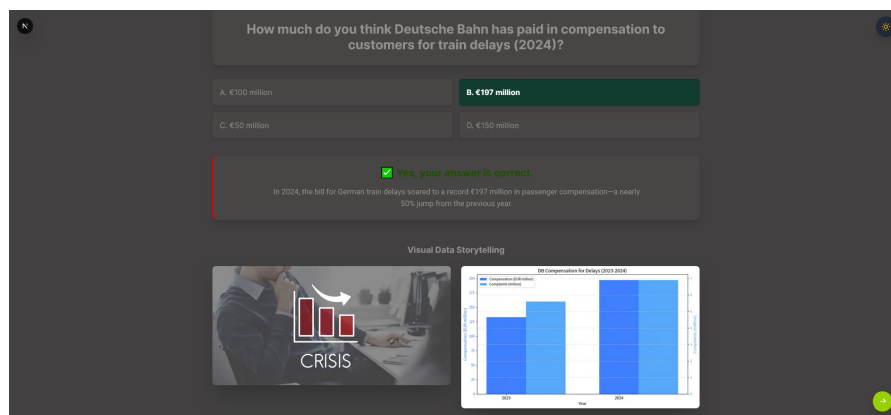


Figure 4.15: Dashboard - Interactive Quiz [27]

- **Dynamic Data Visualizations:** Each tab integrates various charts, graphs, and factual callouts that update based on user interaction.

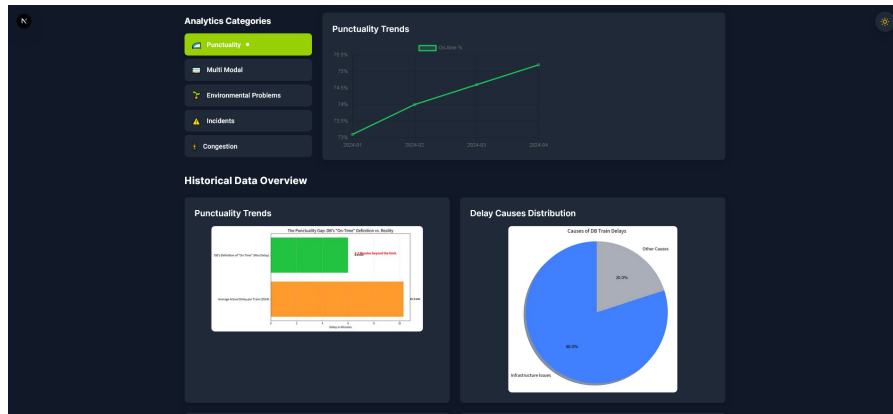


Figure 4.16: Dashboard - Interactive Dashboard [27]

USABILITY EVALUATION

5.1 PREPARATION

5.1.1 *Definition of the Test Goals and Methods*

The primary goal of the usability test was to evaluate the efficacy of the interactive dashboard [27] as a medium for data storytelling. Specifically, the test aimed to assess:

- **Understanding of the data presented:** Could users extract the key insights regarding mobility in Germany?
- **Perceived clarity:** Was the interface intuitive and the narrative structure logical?
- **Motivation to engage (Immersion):** Did the interactive elements (quiz, map) encourage deeper exploration?
- **Perceived storytelling elements:** Did users recognize the narrative thread across the five thematic tabs?

To achieve these goals, a mixed-methods approach was employed, combining:

- **Thinking Aloud:** Participants were asked to verbalize their thoughts during the interaction to provide insight into their cognitive process.
- **Observation:** A moderator and an observer monitored the session to note friction points and user behavior.
- **Interview:** Semi-structured interviews were conducted pre- and post-test to gauge changes in perception.
- **User Experience Questionnaire (UEQ):** [37] [43] This standardized tool was selected to measure user experience across six dimensions (Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty). The benefit of using the UEQ is its ability to provide a benchmark comparison against established successful products.

5.1.2 *Formulation of the Test Tasks and Explanation of the Test Procedure*

The core procedure was designed to simulate a natural discovery process.

- **Thinking Aloud Protocol:** Participants were briefed on the method and encouraged to speak freely about what they were seeing, doing, and feeling.
- **Task - Explore the Dashboard:** Instead of rigid step-by-step instructions, participants were given a 10-minute brainstorming and exploration task. They were asked to freely navigate the five data stories (Punctuality, Congestion, Incidents, Multi-modal, Environment), use the interactive map, and attempt the quiz. [27]

5.1.3 *Selection of the Test Subjects*

The test involved **7 participants**. **Screening Criteria:** No specific exclusion criteria were applied regarding technical expertise. Since the dashboard is intended for the general public (commuters, citizens interested in mobility), a broad selection of participants was appropriate to ensure the tool is usable by non-experts.

5.1.4 *Selection of the Test Environment and Test Object*

The evaluation was conducted in a controlled environment to ensure consistent conditions:

- **Location:** Usability Lab (K3009).
- **Equipment:** The setup included a dedicated monitor for the dashboard [27], microphones for audio recording, and two cameras to capture facial expressions and interaction.
- **Roles:** One project member acted as the Moderator inside the lab, while a second member acted as the Observer in the recording room.

While initial designs were created in Figma [14], the actual test object was the High-Fidelity Functional Prototype [27].

- **Format:** A fully interactive web-based dashboard built with React, Next.js, and Leaflet [27].
- **Reasoning:** A functional code prototype was necessary to genuinely test the dynamic interactions (map filtering, quiz logic) which static prototypes could not emulate.

5.1.5 Creating the Tester Documents

Prior to testing, a suite of documents was prepared to standardize the process:

- **Consent Form:** A formal agreement for audio and video recording and data storage.
- **Questionnaires:**
 - **Pre-Test:** Questions regarding demographics and prior experience with dashboards/data storytelling.
 - **Main Questions:** “How did you perceive the dashboard?”, “What was good?”, “Was the storytelling clear?”
 - **UEQ:** The standard UEQ sheet for quantitative scoring.

ID:	Start:	End:
Introduction: - Welcoming the participant and explaining the purpose of the usability test - There are no right or wrong answers - Your feedback is appreciated. - Explain your recording setup, and why you want - Have the participant sign the consent form for both audio and video recordings.		Notes:
Open questions (3 minutes): - What comes to your mind when you hear the word “Dashboard”? - Have you ever used a dashboard before? If yes, which ones? - o If yes: ▪ How would you describe a dashboard to someone who's never used one before? ▪ What do you expect to see in a dashboard? ▪ For what purpose, in your opinion, are dashboards useful? - o If no: Explanation, what a Dashboard is		Notes:
Topic Relevance - Deutsche Bahn (2 minutes) - How often do you use Deutsche Bahn, and how relevant train delays are to you personally? - Have you ever experienced train delays or compensation claims? - Do you think visualizing mobility data is useful?		Notes:
Main task (5-10 minutes) - Task: Explore on your own - Thinking Aloud: “We would like to hear your thoughts. Please think aloud as you go through the task. Feel free to express your reactions: whether you're pleased, frustrated, or surprised. What are you thinking right now? Why did you just do that? Afterwards, there will be a short evaluation and an interview.” - Let the user navigate the dashboard, DON'T Help unless they are stuck or ask for help)		Notes:
UEQ and Rating: - Giving the UEQ form to the participant and having it filled out - How would you rate the relevance of this topic now? (On a scale of 1 to 5, where 1 is not relevant and 5 is very relevant) - Give an overall rating from 1 to 5 (1 = very poor, 5 = very good)		Notes:
Interview (10-15 Minutes) 1. Central Message: In your opinion, what is the main message or central insight conveyed by this data story or dashboard? 2. First Impression: What is the first thing you notice when interacting with the dashboard? 3. Comprehensibility and Narrative Structure: Were you able to identify a clear narrative or storyline? How well could you follow the overall structure of the data story? 4. Interactivity and Navigation: How did you perceive the level of interactivity? Were you able to navigate freely and find the information you needed? In which areas would you have preferred more or less interaction? 5. Design and User Experience: How would you describe your experience with the visual design of the data story? To what extent does the design support your understanding of the content? 6. Overall Evaluation: What aspects did you find most helpful or appealing? Which elements did you find confusing or unhelpful? Is there anything you consider unclear, ambiguous, or missing?		Notes:
Demographics: Have the participant fill out the demographics form		

Figure 5.1: The Usability Test Guide

5.2 IMPLEMENTATION

- *Performance of a Pre-Test:*
To ensure the technical setup and interview script were fully validated, a pre-test was conducted on November 10th, one day before the main study. A single participant underwent the full procedure. This pilot test confirmed that the instructions were clear and the recording equipment was functioning correctly.
- *Performance of the Main Test:*
The main data collection took place on November 11th.
 - **Duration:** The testing ran continuously for 6 hours.
 - **Procedure:** Each of the 7 participants was given a 30 to 40-minute slot. The session flowed smoothly from onboarding to exploration (Thinking Aloud) to the UEQ. [27]
- *Conducting an Interview:*
Following the exploration phase and the UEQ [43] [37], a semi-structured interview was conducted. This included:
 - **Topic-Related Questions:** Participants rated ideas about their daily transportation usage.
 - **Perception Check:** Participants were asked how their view of German mobility might have been altered by the data stories they just explored.
 - **Brainstorming:** A final open discussion allowed users to reflect on their experience.

Start:	End:
Introduction: - Welcoming the participant and explaining the purpose of the usability test - There are no right or wrong answers - Your feedback is appreciated. - Explain the recording setup - Have the participant sign the consent form for both audio and video recordings	Notes:
Open questions (20 minutes): - What comes to your mind when you hear the word "Dashboard"? - What comes to your mind when you hear the word "Data story"? - Have you ever used a dashboard before? If yes, which ones? - If yes: - How would you describe a dashboard to someone who's never used one before? - What do you expect to see in a dashboard? - For what purposes, in your opinion, are dashboards useful? - If no: Explanation, what a Dashboard and Data story is	Notes: → merge specific data / process. → data all about, say lots of information. → Yes, I products, dashboards, presentation, graphics. → I believe everyone knows dashboards. → depend on purpose. → warning.
Topic Relevant - Deutsche Bahn (2 minutes) how often do you use Deutsche Bahn?	Notes: - Never (1) Rarely (few times a year) - Occasionally (once a month) - Frequently (few times a day) - Very Frequently (almost every day)
how relevant is the topic "Public Transport in Germany" generally to you?	Not Relevant (1) Slightly Relevant (2) Moderately Relevant (3) Relevant (4) Very Relevant (5)
how relevant train delays are to you personally?	Not relevant (doesn't impact me, 1) Slightly relevant (minor inconvenience, 2) Moderately relevant (delays affect me sometimes, 3) relevant (delays often cause problems for me, 4) very relevant (delays have a major impact on my daily life, 5)
how you ever experienced train delays or compensation claims?	Notes: → not colors, very strong dark → strong elements.
how you think evaluating mobility data is useful?	Notes:
In task (2-10 minutes) - Task: Explore on your own - Thinking aloud: "We would like to hear your thoughts. Please think aloud as you go through the task. Feel free to ask me any questions, whether you are confused or surprised. What are you thinking right now? Why did you do that?" Afterwards, there will be a short evaluation and an interview" - Let the user navigate the dashboard, DON'T Help unless they are stuck or ask for help)	Notes:
Q and Rating: Give the UEO form to the participant and having it filled out	Notes: The participant can also think. About during using in the UEO but not a must.
how you rate the relevance of this topic now?	Not Relevant (1) Slightly Relevant (2) Moderately Relevant (3) Relevant (4) Very Relevant (5)
any other things from the relevance of this topic now?	Notes: → Yes, I like it → Yes, I like it

Figure 5.2: The Usability Test Implementation pg 1

ILU	Start:	End:
Interview (15-18 Minutes)		
1. Central Message: In your opinion, what is the main message or central insight conveyed by this data story or dashboard? 2. First Impression: What is the first thing you notice when interacting with the dashboard? 3. Comprehensibility and: Were you able to identify a clear narrative or storyline? 4. Narrative Structure: How well could you follow the overall structure of the data story? 5. Interactivity and Navigation: How did you perceive the level of interactivity? Were you able to navigate freely and find the information you needed? In which areas would you have preferred more or less interaction? 6. Design and Understanding: How would you describe your experience with the visual design of the data story? To what extent does the design support your understanding of the content? 7. Overall Evaluation: What aspects did you find most helpful or appealing? Which elements did you find confusing or unhelpful? Is there anything you consider unclear, ambiguous, or missing? 8. Do you think going through the quiz questions help or add to the experience in your understanding of the data story? Did it motivate you to explore the data story further?	Notes: 1. surprised to learn facts, aimed to be something in a really interactive way. 2. intuitive, easy navigation, also sections of the page. 3. data had different periods, can diff. delays of data. 4. (4-5) (bottom), intuitive, depends in data should be mentioned, more details to each chart. 5. Good, charts are presented elegant way. Good Dashboard should be presented better, more different things, etc. 6. (4-5), very good, missing parts: styling, font color, analytics category must be changed. 7. missing: some data, covering improved, charts must be added. helpful: questions & advancement of current state to previous. 8. surprised, while seeing the facts & figures. it is helpful & potential.	
Demographics: Have the participant fill out the demographics form		

Figure 5.3: The Usability Test Implementation pg 2

5.3 INTERVIEW QUESTIONS AND PROTOCOL

To ensure consistency and depth in qualitative data collection, a standardized interview guide was followed. The questions were categorized into distinct phases:

INITIAL QUESTIONS (PRE-INTERACTION) These questions aimed to establish a baseline of the user's prior knowledge and habits.

- How often do you encounter data dashboards in your daily life (e.g., fitness apps, weather apps)?
- Are you familiar with the concept of "data storytelling"? If so, what does it mean to you?
- Can you describe a recent experience where you used data to make a decision?

TOPIC-RELATED QUESTIONS (MOBILITY CONTEXT) These questions focused on the user's relationship with the subject matter to contextualize their reactions to the dashboard data.

- On a scale of 1 to 10, how would you rate your current satisfaction with public transportation in Germany?
- What are your primary concerns when using public transport (e.g., punctuality, cost, environment)?
- How often do you check mobility-related data (e.g., delay times, congestion maps) before traveling?

MAIN INTERVIEW QUESTIONS (POST-INTERACTION) After the user explored the dashboard and completed the tasks, these reflective questions gathered qualitative feedback on the design and storytelling.

- **Overall Perception:** How did you perceive the dashboard overall? What was your first impression?
- **Strengths:** What aspects of the dashboard worked well for you? (e.g., clarity, navigation, aesthetics)
- **Weaknesses:** What areas do you think need improvement? Were there any points of confusion?
- **Storytelling Clarity:** Was the data story clear to you? Did the sequence of information make sense?
- **Quiz Engagement:** How did the interactive quiz influence your engagement with the data? Did it help you learn something new?
- **Visualizations:** Were the charts and maps easy to understand? Did they effectively communicate the key insights?
- **Navigation:** Did you find it easy to move between the different sections (e.g., from the quiz to the map)?
- **Impact:** Has exploring this dashboard altered your perception of mobility issues in Germany? If so, how?

CLOSING AND DEMOGRAPHICS The session concluded with specific demographic questions to profile the participant group as outlined in the usability testing protocol:

- **Age**
- **Gender**
- **Job / Occupation**
- **Degree (Highest level of education)**

5.4 EVALUATION OF THE TEST RESULTS

5.4.1 *Quantitative Evaluation Analysis: UEQ [43] [37]*

The analysis of the UEQ [43] [37] data yielded highly positive results, confirming the efficacy of the prototype's design and implementation. The User Experience Questionnaire (UEQ) measures user experience across 26 items which are aggregated into six scales. These scales cover both pragmatic quality (Perspicuity, Efficiency, Dependability) and hedonic quality (Stimulation, Novelty), as well as a general impression of Attractiveness.

The measured scale means were set in relation to existing values from the UEQ Benchmark Data Set [43] [37]. This benchmark contains data from 21,175 persons across 468 studies concerning different products (business software, web pages, web shops, social networks). The comparison allows for conclusions about the relative quality of the evaluated dashboard compared to other established interactive products.

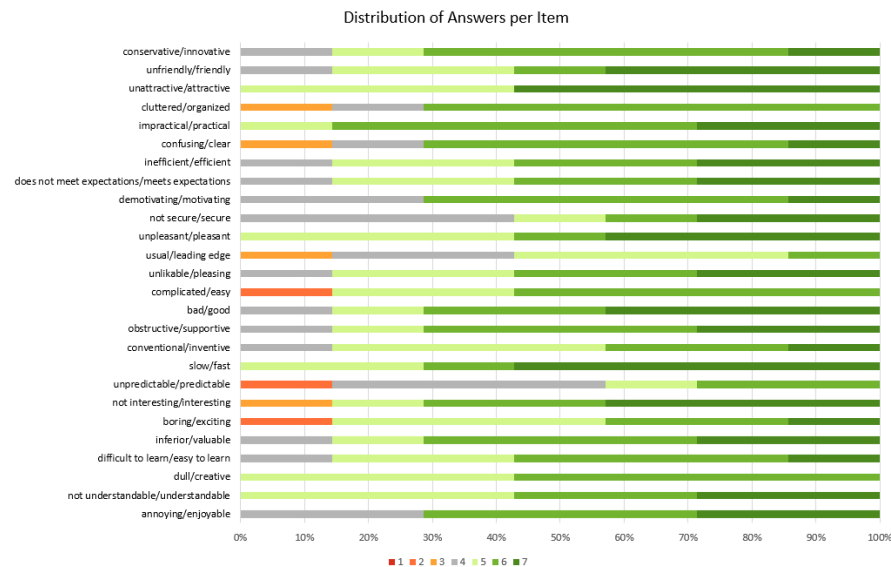


Figure 5.4: Calculated Percentage of answered UEQ [37]

Based on the analysis of the 7 participants, the dashboard achieved the following results:

- **Attractiveness (Mean: 1.90) - “Excellent”:** This scale represents the overall impression of the product. An “Excellent” rating places the dashboard in the top 10% of results in the benchmark. This indicates that users found the dashboard visually appealing and generally enjoyable to use.
- **Perspicuity (Mean: 1.50) - “Above Average”:** This measures how easy it is to understand and learn how to use the product. A rating of “Above Average” (25% of results better, 50% worse) suggests the interface was intuitive for most users, though there may be minor room for improvement in onboarding or clarity.
- **Efficiency (Mean: 1.86) - “Good”:** Efficiency refers to whether users can solve their tasks without unnecessary effort. A “Good” rating (10% of results better, 75% worse) confirms that the dashboard design allowed users to find information and navigate quickly without friction.
- **Dependability (Mean: 1.32) - “Above Average”:** This scale assesses whether the user feels in control of the interaction and

if the system is predictable. The "Above Average" score indicates that, while the system was generally stable and consistent, some users may have experienced minor navigation or feedback uncertainties.

- **Stimulation (Mean: 1.61) - "Good":** Stimulation measures how exciting and motivating it is to use the product. This is a crucial metric for data storytelling. A "Good" score suggests that the interactive elements (such as the quiz and map) successfully engaged users and motivated them to explore the data further.
- **Novelty (Mean: 1.32) - "Good":** This dimension evaluates the creativity and innovation of the design. A "Good" rating indicates that users perceived the dashboard as modern and distinct from standard, static data reports.

Scale	Lower Border	Bad	Below Average	Above Average	Good	Excellent	Mean
Attractiveness	-1.00	0.69	0.49	0.4	0.26	0.66	1.90
Perspicuity	-1.00	0.72	0.48	0.53	0.27	0.5	1.50
Efficiency	-1.00	0.6	0.45	0.45	0.38	0.62	1.86
Dependability	-1.00	0.78	0.36	0.34	0.22	0.8	1.32
Stimulation	-1.00	0.5	0.5	0.35	0.35	0.8	1.61
Novelty	-1.00	0.16	0.54	0.42	0.48	0.9	1.32

Figure 5.5: Mean values for the six user experience scales [37]

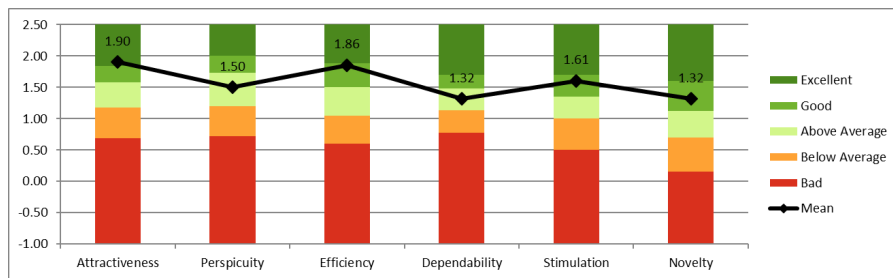


Figure 5.6: Overview of UEQ Results: Mean values for the six user experience scales [37]

5.4.2 Qualitative Evaluation: Analysis of Thinking Aloud and Interviews

To provide a deeper understanding of the user experience, the feedback collected from the semi-structured interviews was analyzed question by question. This granular analysis highlights specific user sentiments regarding the storytelling, design, and interactivity.

1. ANALYSIS OF RATING SCALES The following table summarizes the key ratings collected during the interview protocol. These values represent the average or representative scores given by the participants (N=7).

Question Category	Observed Rating	Participant Feedback Context
Frequency of DB Usage	4 (Frequently)	Participants identified largely as regular commuters who use the service often.
Topic Relevance (General)	5 (Very Relevant)	The topic of public transport was rated as highly significant to general life in Germany.
Topic Relevance (Personal)	6 (Very Relevant)	Train delays were noted as personally critical, with one user noting, "If we miss a connection, we suffer."
Interaction Quality	4.5 (High)	The interaction was rated better than expected ("4.5"), described as "intuitive" and smooth.

Table 5.1: Participant Ratings on Topic Relevance and Interaction

2. PERCEPTION OF THE DATA STORY When asked about their general perception, participants expressed a positive emotional response. Notes from the sessions indicate users were "surprised to learn about the facts" and "amused to see something in a really interactive way." This suggests the dashboard successfully transformed dry statistics into an engaging narrative that captured user attention immediately.

3. STRUCTURE AND LOGIC Participants consistently described the narrative flow as "intuitive." Specific feedback highlighted that the "easy navigation" and the clear division of "sections of the page" helped them understand the logical progression of the story. The breakdown of data into "different periods" was also noted as a positive structural element that aided comprehension.

4. RELEVANCE OF DATA The selected metrics (Punctuality, Compensation) resonated strongly with the participants' personal experiences. Feedback confirmed that the data on "delays" and "compensation" had high "relevancy" to their daily lives, validating the content strategy of focusing on commuter pain points.

5. INTERACTION AND USABILITY On a rating scale (approximated from qualitative comments), the interaction was rated highly (e.g., "4.5"). Users found the system "intuitive," but specific usability issues were identified. One participant noted that "legends in data" needed better integration with the charts (e.g., "circle chart"). While the overall interaction was smooth, these micro-interactions presented minor friction points.

6. FREEDOM OF EXPLORATION Participants appreciated the “elegant way” charts were presented and felt they had sufficient freedom to explore. However, suggestions for improvement included adding “more different Quiz questions” to extend the engagement and refining the “UI potential” to make the dashboard sections even more self-explanatory (“Dashboard should be presented themselves”).

7. VISUAL DESIGN The visual design was rated as “very good” overall. However, critical feedback was provided regarding specific styling choices. Participants pointed out “missing parts” in the design polish, specifically mentioning that “font color” needed adjustment for better readability and that the “analytics category” labels required changing to be more descriptive.

8. ROLE OF THE QUIZ The quiz was unanimously viewed as a “helpful” element. Participants noted it was effective in showing the “advancement of the current state to previous” years. The element of surprise (“Surprised while seeing...”) played a key role in motivating them to explore the dashboard further, confirming the hypothesis that an interactive hook increases user engagement.

5.5 RECOMMENDATIONS AND REFINEMENT

Based on the results from the focus groups and usability tests, several specific improvements were identified to refine the dashboard. The feedback was primarily focused on clarity, design, and narrative effectiveness.

- **Improving Clarity and Simplicity:** Users often experienced that certain views contained excessive information, resulting in visual overload. To address this:
 - **Reduce Clutter:** Simplify complex views to highlight essential data points.
 - **Progressive Disclosure:** Show only important information initially, allowing users to expand for additional details.
 - **Visual Consistency:** Use clear headings and consistent color coding to aid mental modeling.
- **Strengthening Storytelling:** Participants expressed a preference for dashboards that guide them through the information concisely. Recommendations include:
 - **Contextual Descriptions:** Add explanations under charts and figures to clarify what is being shown and why it matters.
 - **Annotations:** Use highlighters and annotations to draw attention to key trends and changes.
 - **Clear Overview:** Provide a strong introductory overview so users know where to begin their journey.
- **Improving Navigation:** Some users experienced navigation confusion, disrupting the user experience. [16]
 - **Menu Structure:** Implement a simple, consistent navigation menu.
 - **Search Functionality:** Include a search bar for quick access to specific metrics or locations.
 - **Sectioning:** Clearly define sections such as “Overview,” “Trends,” and “Details.”
- **Timeline and Interaction** Feedback indicated that certain dynamic elements required better temporal controls to enhance user understanding.
 - **Interactive Timeline:** Implement a slider or timeline feature that allows users to see how data evolves over time (e.g., tracking delays over a year).
 - **Comparison Mode:** Enable users to compare different time periods or regions side-by-side.
 - **Hover Details:** Refine hover interactions to display more granular data on demand without cluttering the main view.

5.6 FUTURE WORK: LIVE DATA INTEGRATION

A critical next step for the project is the transition from static datasets to real-time data integration. In future iterations, live data from Deutsche Bahn (DB) [27] will be implemented across all metrics.

Currently, the dashboard relies on static snapshots for demonstration purposes. The integration of live APIs [16] [35] will allow for:

- **Real-Time Interactive Map:** The map will dynamically visualize which areas are currently experiencing high punctuality versus those suffering from congestion. [16]
- **User-Centric Navigation:** Features will be added to allow users to locate the nearest *Bahnhof* (train station) or *Busbahnhof* (bus station) based on their current location, facilitating immediate travel planning. [16]
- **Dynamic Metrics:** All charts and graphs regarding punctuality, incidents, and environmental impact will update in real-time to reflect the current state of the German mobility network. [35]
- **Dynamic Quiz Updates:** To maintain user engagement over time, the quiz content will be programmed to update automatically based on current trends and news. Regularly changing the questions ensures that returning users are constantly challenged with fresh content, increasing the likelihood of repeat visits and sustained interest in the platform.

5.7 LIMITATIONS

While the usability test provided valuable insights, the study design inherently includes certain limitations that should be considered when interpreting the results.

- **Sample Size:** The UEQ is optimally designed for a larger number of participants (typically $N > 20$) to achieve statistical significance. With a sample size of $N=7$, the quantitative results should be interpreted as indicative trends rather than definitive statistical proof. However, for qualitative usability testing, 5-7 users are often sufficient to identify the majority of usability problems.
Solution: Conduct supplementary A/B testing or remote unmoderated testing with a larger, more diverse user base to validate quantitative trends.
- **Participant Selection Bias:** The participants were recruited from a university environment, which may bias the sample toward individuals with higher digital literacy and familiarity with academic or data-driven interfaces. This group might navigate complex dashboards more easily than the general population.
Solution: Recruit future participants from varied demographics

(e.g., non-academic backgrounds, different age groups) using public forums or transit stations to ensure a representative sample.

- **Prior Knowledge:** Participants' prior experience with public transport in Germany varied. Those with extensive experience ("frequent travelers") might have approached the data with preconceived notions or frustrations, potentially influencing their "Relevance" and "Stimulation" scores differently than users who sometimes use public transport.

Solution: Segment data analysis by user expertise (novice vs. expert) to identify specific needs and tailor the interface for both user types.

- **Test Environment:** The controlled lab setting, while ensuring consistency, does not perfectly replicate the chaotic, distracted environments (e.g., on a moving train or busy platform) where users would typically access a mobility dashboard.

Solution: Perform field studies where participants use the dashboard in real-world settings, such as on a train or at a busy station, to assess usability under distraction.

SUMMARY

This study explored how data-driven storytelling principles can enhance the design, usability, and overall user experience of mobility dashboards. The aim was to understand how users perceive visual information, what makes dashboards helpful or frustrating in understanding a story, and how narrative elements can help improve the communication of complex data. A mobility dashboard becomes a meaningful data story when it goes beyond just numbers, charts, and visualizations. It integrates narrative structure and explanatory context, helping the user interpret the data and motivating them to engage and explore further.

The methodology combined qualitative and quantitative approaches in an iterative user-centered design process. Two focus group sessions were conducted to explore early perceptions of existing dashboards and collect expectations, preferences, and pain points. The participants reviewed two contrasting dashboards, Belgian Mobility Dashboard [3] and German Klima Dashboard [24] to provoke discussion about layout, interactivity, information density, and narrative flow. The second phase applied their feedback to refine a prototype dashboard [14], which was then evaluated in individual usability tests. This method allowed for triangulation between exploratory group-based insights and more structured, task-based performance evaluation. The overall methodological approach ensured that user feedback informed design decisions at multiple stages of development.

The focus groups revealed that users strongly value clarity, simplicity, and a clear narrative structure. During both sessions, participants expressed that a dashboard should allow them to quickly understand the “story” behind the data without having to interpret overly dense visualizations. The Belgian Mobility Dashboard [3] was praised for its clarity and compact communication style. In contrast, the Klima Dashboard [24] was appreciated for its interactivity but criticized in both groups for visual overload or missing storytelling elements such as timelines, annotations, and search functionality. Participants also sketched their own ideal dashboard designs, revealing key themes: a desire for customizable views, the importance of temporal context, a preference for minimalist layouts, and the need for accessible explanations of charts. These insights provided concrete direction for refining the prototype, helping to prioritize features and address common pain points before usability testing.

Usability tests evaluated how well the refined prototype supported task completion, comprehension, and user satisfaction. Participants completed realistic tasks, and their performance, comments, and UEQ scores were analyzed [43] [37]. The results indicated that the integration of storytelling principles such as a clear narrative structure,

improved annotation, and a more intuitive layout led to improved comprehension and smoother navigation. However, challenges remained, including issues with chart detail, loading speed, and the need for clearer explanations of complex indicators. The usability tests confirmed that while the revised design addressed many initial concerns raised in the focus groups, iterative refinement would still be necessary to achieve a fully polished user experience.

APPENDIX

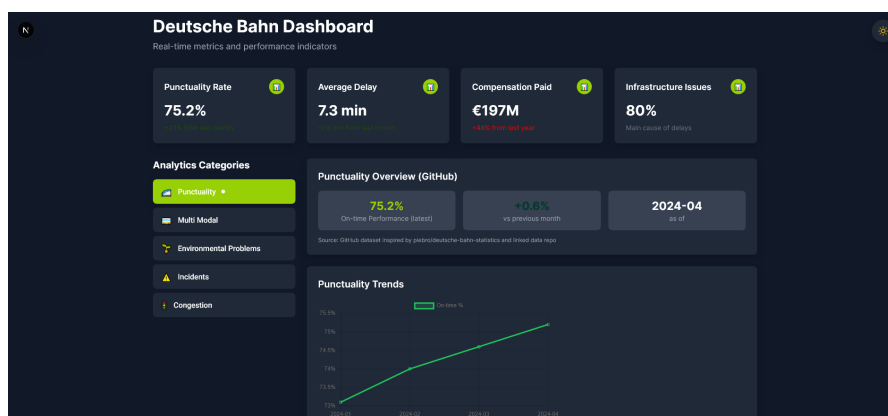


Figure A.1: Dashboard [27]

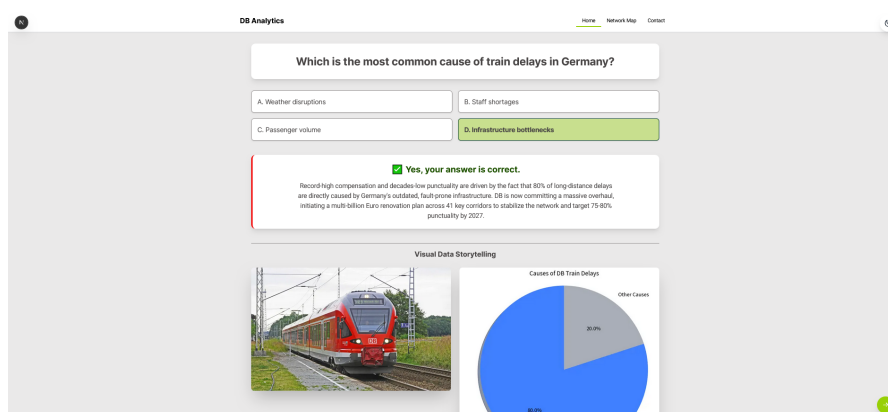


Figure A.2: Dashboard - Quiz Question 2 [27]

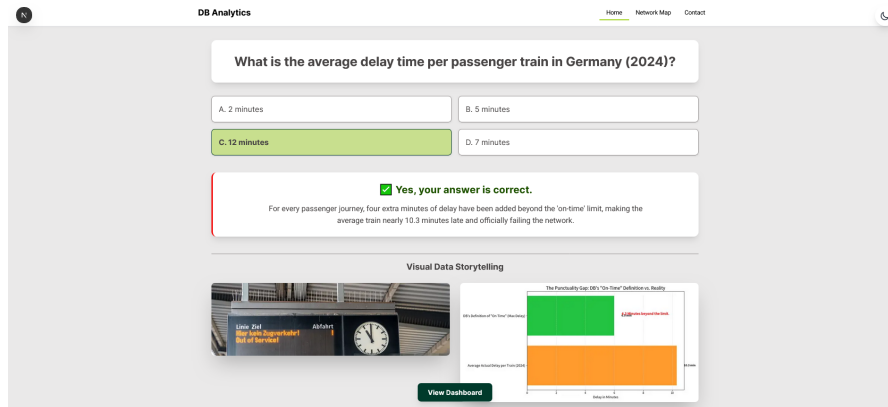


Figure A.3: Dashboard - VIEW DASHBOARD [27]

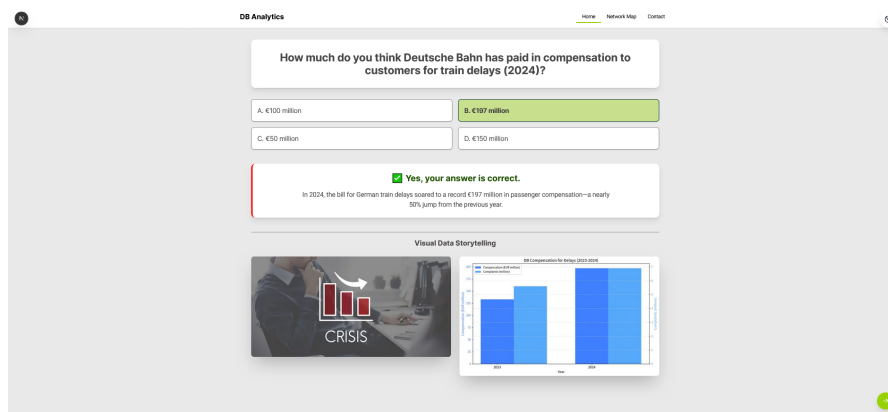


Figure A.4: Dashboard - Quiz Question 1 [27]

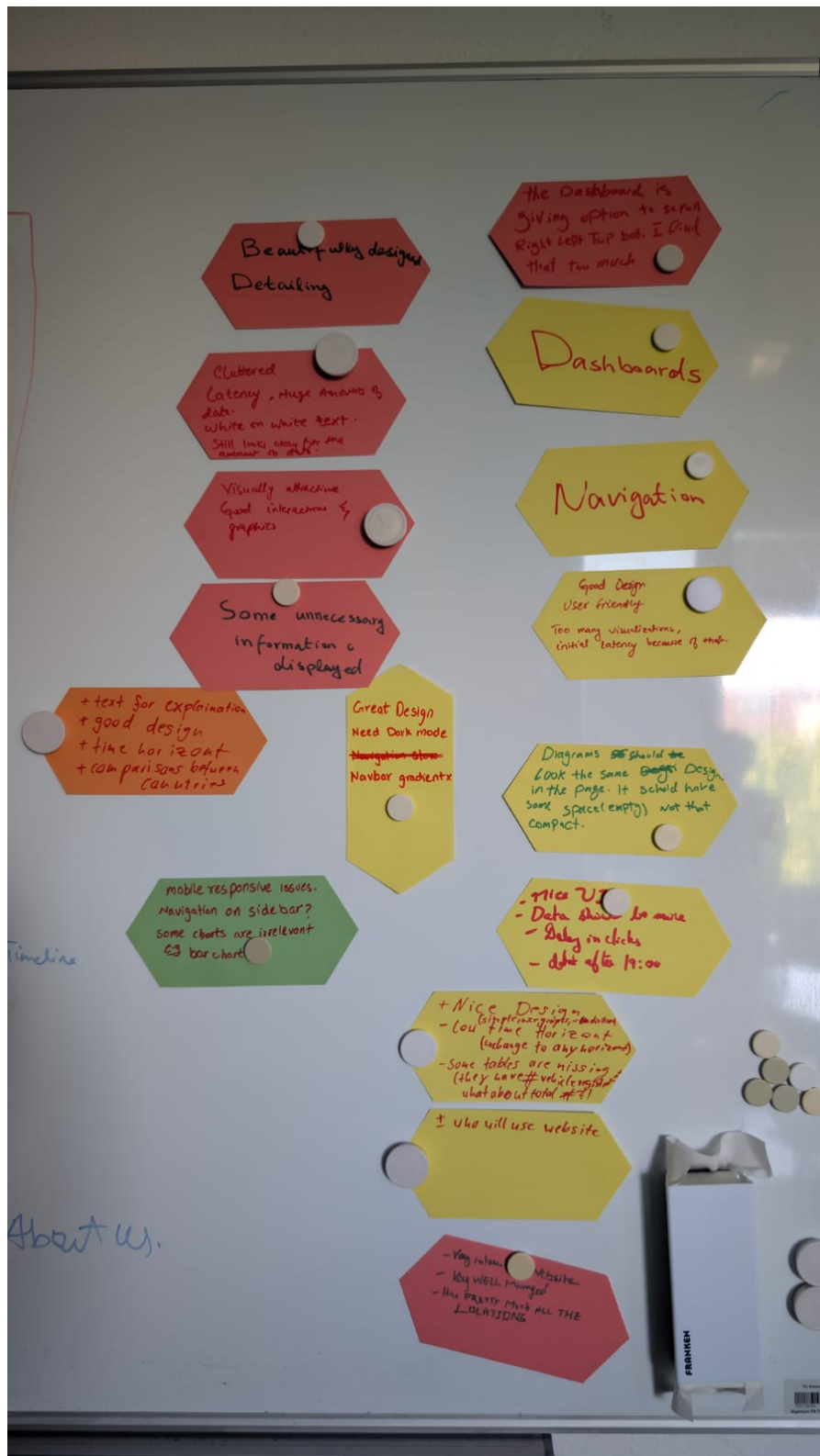


Figure A.5: Use of cards by the participants in Focus Group Session 1

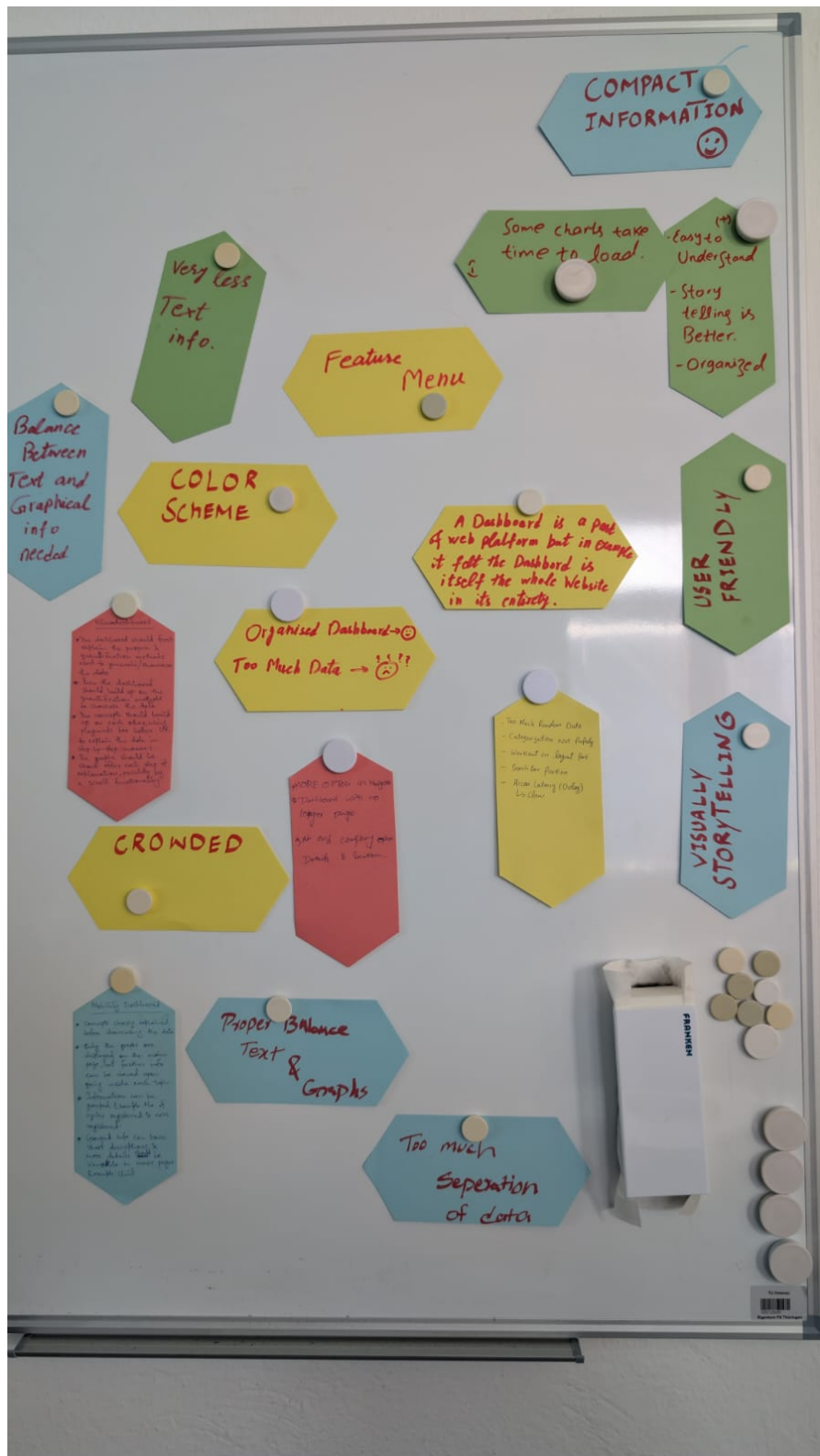


Figure A.6: Use of cards by the participants in Focus Group Session 2

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DECLARATION

This work is a collective work. Mr. Jawad Munir wrote and edited the following sections.

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- 6

Mr. Muhammad Behzad Hussain wrote and edited the following sections.

- 4
- 5

Mr. Muhammad Ali wrote and edited the following sections.

- 1
- 2

We hereby declare that we made limited use of AI tools, such as ChatGPT and Gemini, for the refinement of language and structure in chapters Chapters 1,2 and 4 of this work. The support provided by AI was solely aimed at improving clarity and expression, without replacing our own research or critical thinking.

All sources, literature, and tools that we have used are properly cited and acknowledged.

I hereby declare that this thesis was created autonomously without using other than the stated references. All parts which are cited directly or indirectly are marked as such. This thesis has not been used in the same or similar forms in parts or total in other examinations.

Ilmenau, 30.11.2025



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